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Tumors mimic the niche to inhibit neighboring stem cell differentiation

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This study presents results supporting a model that tumorous germline stem cells (GSCs) in the *Drosophila* ovary mimic the stem cell niche and inhibit the differentiation of neighboring cells. The **valuable** findings show that GSC tumors often contain non-mutant cells whose differentiation is suppressed by the GSC tumorous cells. However, the evidence showing that the GSC tumors produce BMP ligands to suppress differentiation of non-mutant cells is **incomplete** due to concerns about the new HCR data.

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Abstract

Although it is well-established that stem cells maintain tissue homeostasis while tumors disrupt it, the mechanisms by which tumors influence the development of nearby stem cells remain poorly understood. Using *Drosophila* ovaries as a model system, here we discovered that *bam* or *bgn* mutant germline tumors inhibit the differentiation of neighboring wild-type germline stem cells (GSCs). Mechanistically, these tumor cells mimic the stem cell niche by secreting the BMP ligands Dpp and Gbb, but at reduced levels, resulting in moderate BMP signaling activation in adjacent GSCs. Such BMP signaling activation is sufficient to repress *bam* transcription, thereby blocking GSC differentiation. To our knowledge, this is the first example that tumors can functionally mimic a stem cell niche to inhibit the differentiation of neighboring wild-type stem cells. Similar regulatory paradigms may operate in mammalian tissues, including humans, during tumorigenesis.

Introduction

The homeostasis of many tissues in our bodies is maintained by adult stem cells, but this balance can be disrupted by tumor cells. What occurs when tumorigenesis intersects with stem cell development? To address this question, a mosaic-analysis model system is essential, where wild-type stem cells develop alongside tumor cells. *Drosophila* offers an exceptional model for such studies, as it allows for the efficient generation of mosaic clones through various established methods (Germani et al., 2018 [↗](#); Pastor-Pareja and Xu, 2013 [↗](#)).

In *Drosophila* ovaries, germline stem cells (GSCs) play a crucial role in sustaining normal oogenesis and maintaining fertility (Fuller and Spradling, 2007 [↗](#); Lin, 1997 [↗](#)). These GSCs reside in a specialized microenvironment known as the stem cell niche (hereafter referred to as niche) (Xie and Spradling, 2000 [↗](#)). Typically, a GSC undergoes asymmetric division, generating two distinct daughter cells: one remains in the niche to self-renew as a GSC, while the other, called a cystoblast, exits the niche and initiates differentiation. During the differentiation process, each cystoblast performs exactly four rounds of mitotic division with incomplete cytokinesis to produce 16 interconnected cystocytes, forming a germline cyst. In each germline cyst, only one germ cell is destined to become the oocyte, while the remaining 15 differentiate into nurse cells that support the development of the oocyte (Figure 1A [↗](#)) (Fuller and Spradling, 2007 [↗](#); Lin, 1997 [↗](#)). The principal niche signals are bone morphogenetic protein (BMP) ligands, including

Decapentaplegic (Dpp) and Glass bottom boat (Gbb), which are secreted by terminal filament (TF) and cap cells (Chen and McKearin, 2003a [↗](#); Li et al., 2016 [↗](#); Song et al., 2004 [↗](#); Xie and Spradling, 1998 [↗](#), 2000 [↗](#)). These ligands activate BMP signaling in GSCs, leading to the transcriptional repression of *bag of marbles* (*bam*), a key gene that promotes differentiation. In contrast, BMP signaling is inactive in cystoblasts, allowing Bam to be expressed and drive their differentiation (Chen and McKearin, 2003a [↗](#); Song et al., 2004 [↗](#)). Bam carries out this function in collaboration with its partner, Benign gonial cell neoplasm (BgcN) (Li et al., 2009 [↗](#); Ohlstein et al., 2000 [↗](#)).

GSCs mutant for *bam* or *bgcN* fail to differentiate and instead hyper-proliferate, forming a well-established *Drosophila* germline tumor model (Lavoie et al., 1999 [↗](#); McKearin and Ohlstein, 1995 [↗](#); McKearin and Spradling, 1990 [↗](#); Niki and Mahowald, 2003 [↗](#)). Notably, these germline tumor cells competitively displace wild-type GSCs from the niche (Jin et al., 2008 [↗](#)). The resulting displacement creates a microenvironment where wild-type GSCs are surrounded by tumor cells, providing an excellent model system to study stem cell behavior in tumor neighborhoods.

Here, we demonstrate that *bam* or *bgcN* mutant germline tumors inhibit the differentiation of neighboring wild-type GSCs by functionally mimicking the stem cell niche. This mechanism may be conserved in mammals, including humans, during tumorigenesis, where malignant cells could similarly disrupt normal stem cell development.

Results

Germline tumors inhibit the differentiation of neighboring wild-type GSCs

To generate *bam* or *bgcN* mutant germline clones, we employed either *nos>FLP/FRT* or *hs-FLP/FRT* system that we previously established (Zhang et al., 2023 [↗](#); Zhao et al., 2018 [↗](#)). These two systems induce the expression of FLP recombinase either germline-specifically (*nos-GAL4-VP16/UASz-FLP*) or via heat shock (*hs-FLP*). The expressed FLP recombinase targets the *FRT* sites to mediate mitotic recombination on homologous chromosome arms, generating adjacent GFP-negative (*bam* or *bgcN* mutant) and GFP-positive (wild-type) germ cell populations (Figure 1B [↗](#)). Remarkably, we observed that many wild-type germ cells located outside the niche retained a GSC-like single-germ-cell (SGC) morphology (Figure 1C, D [↗](#)), even when encapsulated within egg chambers (Figure 1—figure supplement 1 [↗](#)). Under normal conditions, GSCs that exit the niche differentiate into interconnected germline cysts, where germ cells are linked rather than remaining as individual, isolated cells (Fuller and Spradling, 2007 [↗](#); Xie and Spradling, 2000 [↗](#)). To rule out the possibility that the SGC phenotype is an artifact caused by GFP expression, we repeated the experiments using RFP and arm-lacZ as alternative mosaic-analysis markers. Consistent results were observed (Figure 1E, F [↗](#)), confirming that the phenotype is not attributable to GFP.

To further confirm that these SGCs exhibit GSC-like characteristics, we conducted anti- α -Spectrin immunofluorescent staining, a method that labels a germline-specific organelle known as the spectrosome in GSCs and cystoblasts, and the fusome in cystocytes. GSCs perform complete cell division, whereas cystocytes undergo incomplete cytokinesis, remaining interconnected through fusomes and ring canals. Consequently, spectrosomes appear as dot-like structures, while fusomes exhibit branched morphologies (Figure 1G [↗](#)) (Lin et al., 1994 [↗](#)). To accurately capture the three-dimensional (3D) architecture of spectrosomes and fusomes, we acquired z-stack images using confocal microscopy. Strikingly, these SGCs displayed dot-like spectrosomes, closely resembling those observed in wild-type GSCs and *bam* or *bgcN* mutant GSC-like tumor cells (Figure 1H, I [↗](#)). We also considered the possibility that SGCs might arise through the dedifferentiation of the cystocytes in germline cysts surrounded by germline tumors. If this were the case, such cystocytes would initially undergo complete cell division, leaving behind midbodies as markers of the late cytokinesis stage. When visualized by anti- α -Spectrin immunofluorescence, midbody appears as a central sphere that is slightly connected to two larger flanking structures, resembling a variant of nunchucks (Mathieu et al., 2022 [↗](#)). Notably, in our analyses of over 50 germline cysts surrounded by *bam* mutant germline tumors, none contained midbodies, suggesting that dedifferentiation is

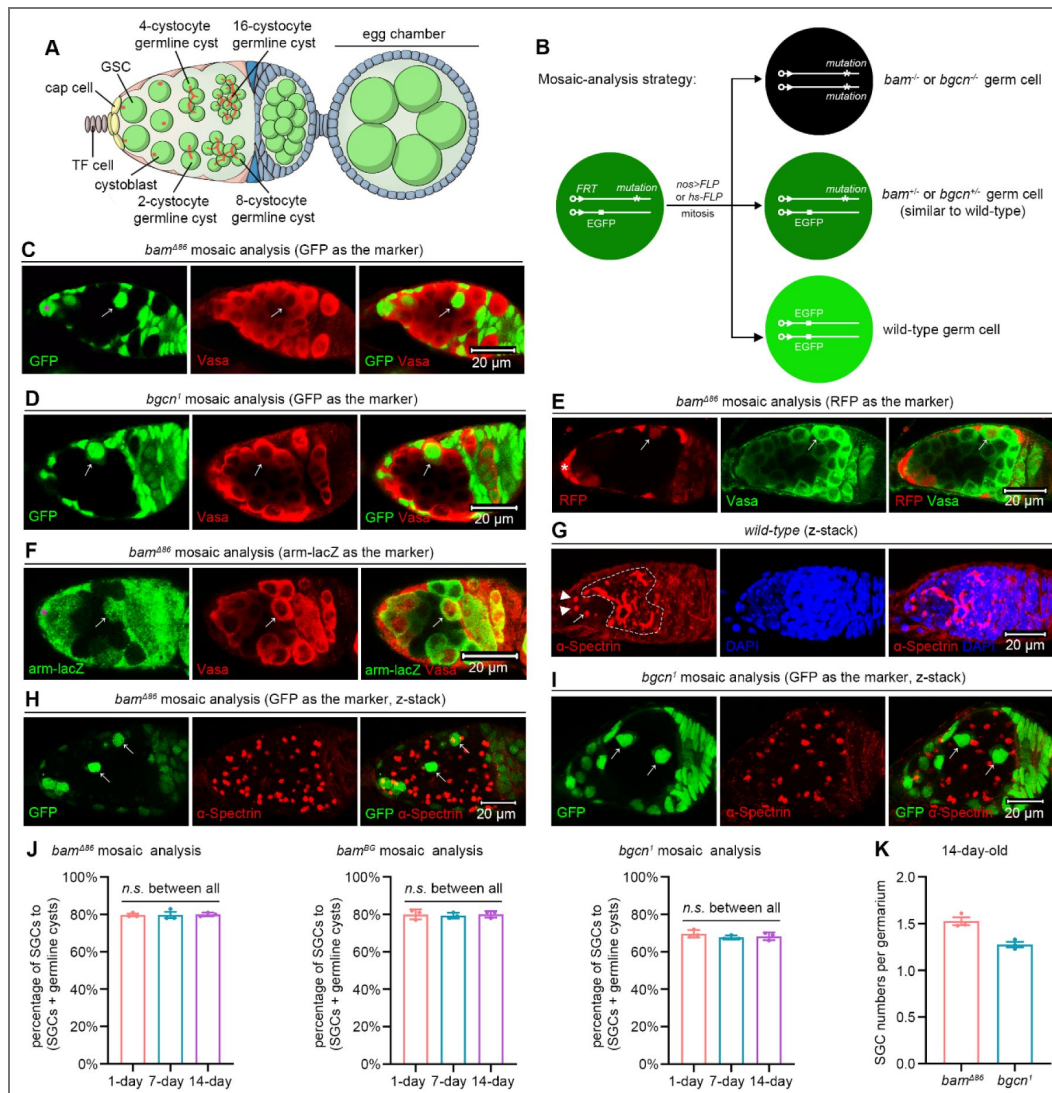


Figure 1. *bam* or *bgcn* mutant germline tumors inhibit the differentiation of neighboring wild-type GSCs.

(A) Schematic cartoon for early oogenesis. The red dots and branches indicate spectrosomes and fusomes, respectively. TF cell: terminal filament cell; GSC: germline stem cell. (B) Mosaic analysis strategy. The FLP recombinase triggers mitotic recombination by targeting *FRT* sequences. The *nos>FLP* method restricts FLP expression to the germline, while the *hs-FLP* method enables heatshock-inducible FLP expression. (C-F) Representative samples. The asterisks mark cap cells, and the arrows indicate SGCs that have exited the niche and are surrounded by *bam* or *bgcn* mutant germline tumors. Vasa, a germ cell marker, should label all germ cells. However, due to poor tumor permeability, staining often fails to detect tumorous germ cells in the central region (see Vasa panels in D-F). (G-I) Representative samples (z-stack projections). In (G), the arrowheads and arrow respectively mark two GSCs and one cystoblast, all containing dot-like spectrosomes, while the dotted lines delineate cystocytes with branched fusomes. In (H) and (I), the arrows denote SGCs that also contain dot-like spectrosomes, akin to GSCs and the adjacent GSC-like tumor cells. (J and K) Quantification data. For each experiment, three independent replicates were performed, and data represent mean $\pm$ SEM. In (J), over 100 SGCs and germline cysts were quantified per replicate, and statistical significance was determined by one-way ANOVA. *n.s.* ($P > 0.05$). In (K), over 100 germaria were quantified per replicate. The online version of this article includes the following source data and figure supplement for figure 1: [Source data 1](#). [All genotypes](#). [Source data 2](#). [Raw quantification data](#). [Figure supplement 1](#). SGCs appear in egg chambers.

unlikely to be the primary mechanism responsible for the SGC phenotype. Together, these findings indicate that *bam* or *bgcn* mutant germline tumors inhibit the differentiation of neighboring wild-type GSCs.

To quantify the SGC phenotype, which requires the presence of both germline tumors and out-of-niche wild-type germ cells, we analyzed germaria containing both. In 14-day-old fly ovaries, 70% of germaria (432/618) met this criterion. We calculated the percentage of SGCs relative to the total number of SGCs and germline cysts, considering the out-of-niche germ cells that are either fully enclosed by germline tumors (e.g., the right SGC in Figure 1I and the marked germline cyst in Figure 2G) or in contact with wild-type germ cells or somatic cells on only one side (e.g., the left SGC in Figure 1I and the outlined germline cyst in Figure 4C). Notably, the SGC phenotype was consistent across the 14-day period analyzed (Figure 1J). For subsequent quantification, we used 14-day-old flies, taking advantage of the robust germline tumor growth at this age to ensure efficient analysis. In qualifying germaria, the average number of SGCs was approximately 1.5 (Figure 1K).

The inhibition of differentiation in SGCs relies on the lack of Bam expression

Given that Bam is the key factor promoting GSC differentiation (McKearin and Ohlstein, 1995; Ohlstein and McKearin, 1997), we were very curious about the expression of Bam in SGCs. At first, we assessed Bam protein levels using immunofluorescent staining with an anti-BamC antibody (McKearin and Ohlstein, 1995). Strikingly, none of the SGCs examined ($n > 100$) were BamC-positive (Figure 2A). Then, we analyzed *bam* transcription levels using a *bamP*-GFP reporter (Chen and McKearin, 2003b). 100% of GSCs within the niche ($n = 153$) were GFP-negative, while 98% of cystoblasts ($n = 106$) were GFP-positive (Figure 2B, C), confirming that *bam* transcription is associated with the initiation of GSC differentiation (McKearin and Ohlstein, 1995). Notably, 74% of SGCs ($n = 132$) were GFP-negative, while the remaining 26% were GFP-positive (Figure 2B, C). This suggests that SGCs can be categorized into two distinct groups: those resembling GSCs (GSC-like) and those resembling cystoblasts (cystoblast-like). The cystoblast-like SGCs may have already initiated their differentiation program toward becoming cystocytes. Since *bam* transcription initiates in cystoblasts (McKearin and Spradling, 1990) but Bam proteins accumulate predominantly in cystocytes (McKearin and Ohlstein, 1995), the Bam protein levels in these cystoblast-like SGCs are likely below the detection threshold at this early stage.

Next, we asked whether ectopic expression of Bam can drive SGCs to differentiate. To address this, we established two experimental scenarios: one with the *hs-bam* element and one without as the control. In the *hs-bam* scenario (with *hs-bam*), GFP-positive germ cells are wild-type (carrying *hs-bam*), while GFP-negative cells are *bam* mutant (lacking *hs-bam*). In the control scenario (without *hs-bam*), GFP-positive cells are wild-type, and GFP-negative cells are *bam* mutant (Figure 2D, see genotypes in Source data 1). To induce ectopic Bam expression, 12-day-old female flies were subjected to heatshock treatment, which involved heating at 37°C for two hours, twice daily with a 6-hour interval, and over two consecutive days. In the absence of heatshock treatment, the percentage of SGCs in ovaries of both genotypes showed no significant difference at either 12 or 14 days (Figure 2E-H), indicating that the *hs-bam* element alone, without heatshock, does not affect the phenotype. However, following heatshock treatment, the percentage of SGCs in ovaries with *hs-bam* was markedly reduced compared to those without *hs-bam* (Figure 2E-H), suggesting that ectopic Bam expression can drive SGCs to differentiate. Collectively, these results support that the differentiation defects of SGCs are due to the lack of Bam expression.

SGCs retain moderate BMP signaling activation

Within the niche, BMP signaling functions to repress *bam* transcription to inhibit GSC differentiation (Chen and McKearin, 2003a; Song et al., 2004). To investigate BMP signaling activation in SGCs, we employed immunofluorescent staining for pMad, a well-characterized marker of BMP signaling activity (Kai and Spradling, 2003). Surprisingly, we observed undetectable pMad levels in all SGCs examined ($n > 100$) (Figure 3A, B). To investigate this

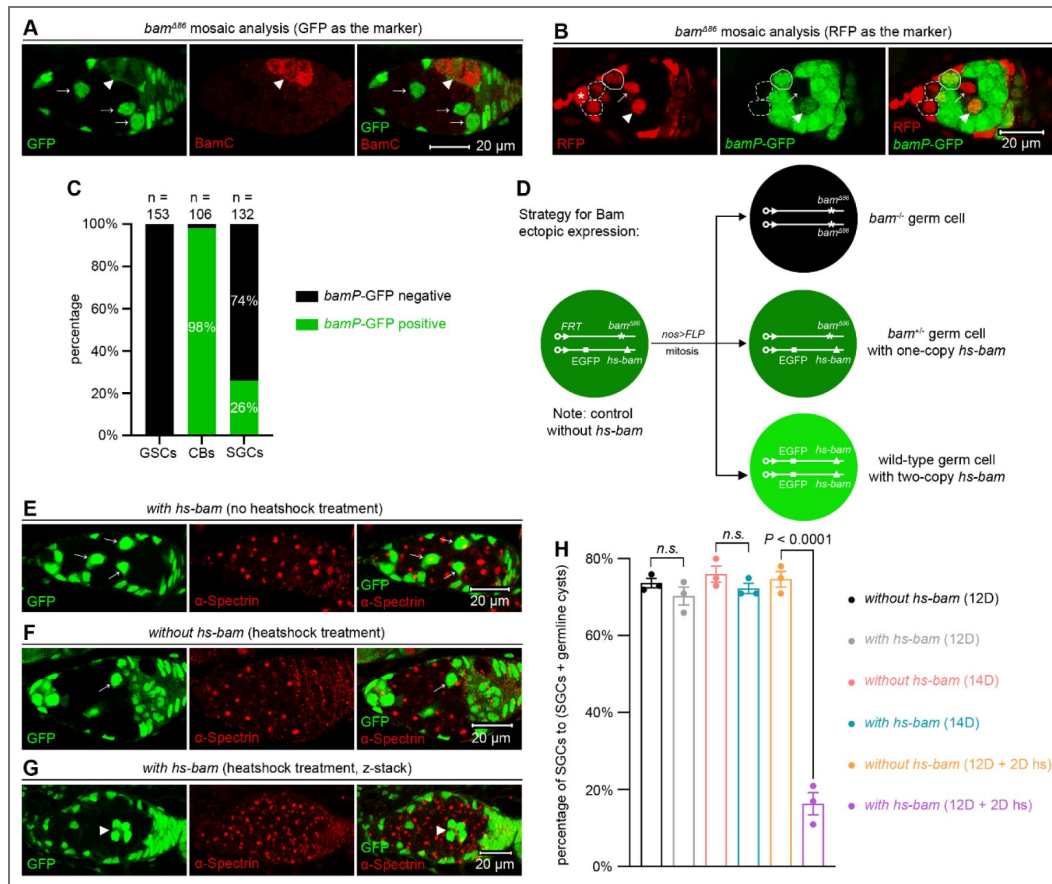


Figure 2. The inhibition of SGC differentiation depends on the lack of Bam expression.

(A) Representative sample. The arrowhead marks a BamC-positive 4-cystocyte germline cyst, while the arrows indicate BamC-negative SGCs. (B) Representative sample. The asterisk denotes cap cells, and the dotted circles outline *bamP*-GFP-negative GSCs. The solid circle marks a *bamP*-GFP-positive cystoblast. The arrow and arrowhead point to *bamP*-GFP-negative and -positive SGCs, respectively. (C) Quantification data. 14-day-old flies were used for the analyses. CBs: cystoblasts. (D) Schematic of the experimental strategy for (E-H). In “*with hs-bam*” flies (E and G), wild-type germ cells (both *bam*^{+/+} and *bam*^{+/-}) carry the *hs-bam* transgene, while control “*without hs-bam*” flies (F) lack this element in their wild-type germ cells. (E-G) Representative samples. The arrows mark SGCs with dot-like spectrosomes, while the arrowhead indicates a 4-cystocyte germline cyst containing branched fusomes. (H) Quantification data. For each experiment, three independent replicates were performed, with over 100 SGCs and germline cysts quantified per replicate. Data represent mean ± SEM, and statistical significance was determined by t test. *n.s.* (*P* > 0.05). The online version of this article includes the following source data for figure 2: Source data 1. All genotypes. Source data 2. Raw quantification data.

further, we examined the activity of *Dad-lacZ*, a highly sensitive BMP signaling reporter known to be activated not only in GSCs but also in cystoblasts (Kai and Spradling, 2003 [\[1\]](#); Song et al., 2004 [\[2\]](#)). Notably, 73% of SGCs were lacZ-positive ($n = 107$), a proportion lower than that of GSCs within the niche, which showed 100% lacZ positivity ($n = 122$) (Figure 3C, D [\[3\]](#)). Furthermore, when comparing *Dad-lacZ* expression levels exclusively in lacZ-positive cells, we found that SGCs exhibited significantly lower expression levels than GSCs within the niche (Figure 3C, E [\[4\]](#)). These findings indicate that BMP signaling is activated in SGCs but at lower levels than those in GSCs within the niche.

Beyond maintaining *Drosophila* female GSCs in the niche, BMP signaling also promotes their division (Xie and Spradling, 1998 [\[5\]](#)). Since the activation levels of BMP signaling in SGCs were lower than those in GSCs within the niche, we hypothesized that SGCs would exhibit slower proliferation rates than GSCs. To test this hypothesis, we performed BrdU incorporation assays. The results revealed that only 4.5% of SGCs were BrdU-positive ($n = 1034$), a significantly lower proportion than the 7.8% observed in GSCs within the niche ($n = 1337$) (Figure 3F-H [\[6\]](#)). These findings further corroborate the reduced activation of BMP signaling in SGCs relative to GSCs.

BMP signaling inhibits SGC differentiation

Then we investigated whether BMP signaling functions to inhibit SGC differentiation. The BMP type II receptor Punt and the co-Smad Medea (Med) are essential for maintaining GSC stemness within the niche (Xie and Spradling, 1998 [\[5\]](#)). To investigate whether they are also required to inhibit SGC differentiation, we established a genetic scenario, in which GFP^{+/+} RFP^{-/-} germ cells are *punt*^{-/-} or *med*^{-/-}; GFP^{+/+} RFP^{+/+} germ cells are *punt*^{+/+}*bam*^{+/+} or *med*^{+/+}*bam*^{+/+} (similar to wild-type); and GFP^{-/-} RFP^{+/+} germ cells are *bam*^{-/-}. In control experiments (with no *punt* or *med* mutation), GFP^{+/+} RFP^{-/-} germ cells are wild-type; GFP^{+/+} RFP^{+/+} germ cells are *bam*^{+/+} (similar to wild-type); and GFP^{-/-} RFP^{+/+} germ cells are *bam*^{-/-} (Figure 4A [\[7\]](#), see genotypes in Source data 1 [\[8\]](#)). Strikingly, the proportion of *punt*^{-/-} or *med*^{-/-} SGCs relative to total SGCs was significantly lower than in controls (Figure 4B-E [\[9\]](#)). Conversely, among *punt*^{-/-} or *med*^{-/-} germ cells meeting our established criteria for SGC phenotype quantification, germline cysts constituted a higher percentage compared to controls (Figure 4F [\[10\]](#)). These results indicate that Punt and Med function to inhibit SGC differentiation.

Mothers against dpp (Mad) is the primary transcription factor of BMP signaling, and it is also essential for GSC maintenance in *Drosophila* ovaries (Xie and Spradling, 1998 [\[5\]](#)). Unlike *punt* and *med*, which reside on the same chromosome arm (3R) as *bam*, *mad* is located on a separate chromosome arm (2L). To investigate whether Mad is required to inhibit SGC differentiation, we established a genetic scenario, in which GFP^{-/-} germ cells are *mad*^{-/-} and GFP^{+/+} germ cells are *bam*^{-/-}. In control experiments (with no *mad* mutation), GFP^{-/-} germ cells are wild-type and GFP^{+/+} germ cells are *bam*^{-/-} (Figure 4G [\[11\]](#), see genotypes in Source data 1 [\[8\]](#)). Notably, *mad* mutation significantly decreased the SGC proportion relative to controls (Figure 4H-J [\[12\]](#)). These results suggest that, like Punt and Med, Mad also plays a crucial role in suppressing SGC differentiation. Together, these findings demonstrate that BMP signaling contributes to inhibiting SGC differentiation, despite at reduced activation levels.

Germline tumors secrete Dpp and Gbb

The formation of a differentiation niche by escort cells is required for GSC differentiation and is known to be disrupted by *bam* mutant germline tumors (Chen et al., 2022 [\[13\]](#); Kirilly et al., 2011 [\[14\]](#)). Although this niche disruption could contribute to the SGC phenotype, an unaddressed question is the source of the BMP ligands (Dpp and Gbb) that maintain BMP signaling activation within SGCs. Guided by the Occam's Razor principle, we focused on *bam* or *bgn* mutant germline tumor cells, as the SGC phenotype is dependent on their presence. To assess the expression of *dpp* and *gbb*, we employed third-generation *in situ* hybridization chain reaction (HCR) (Choi et al., 2018 [\[15\]](#)). Successful detection was confirmed by prominent signal foci in TF and cap cells (Figure 5A, C [\[16\]](#)). To enable quantitative comparison, all experiments and confocal imaging were performed under identical parameters. Signal intensity within *bam* mutant germline tumors and wild-type

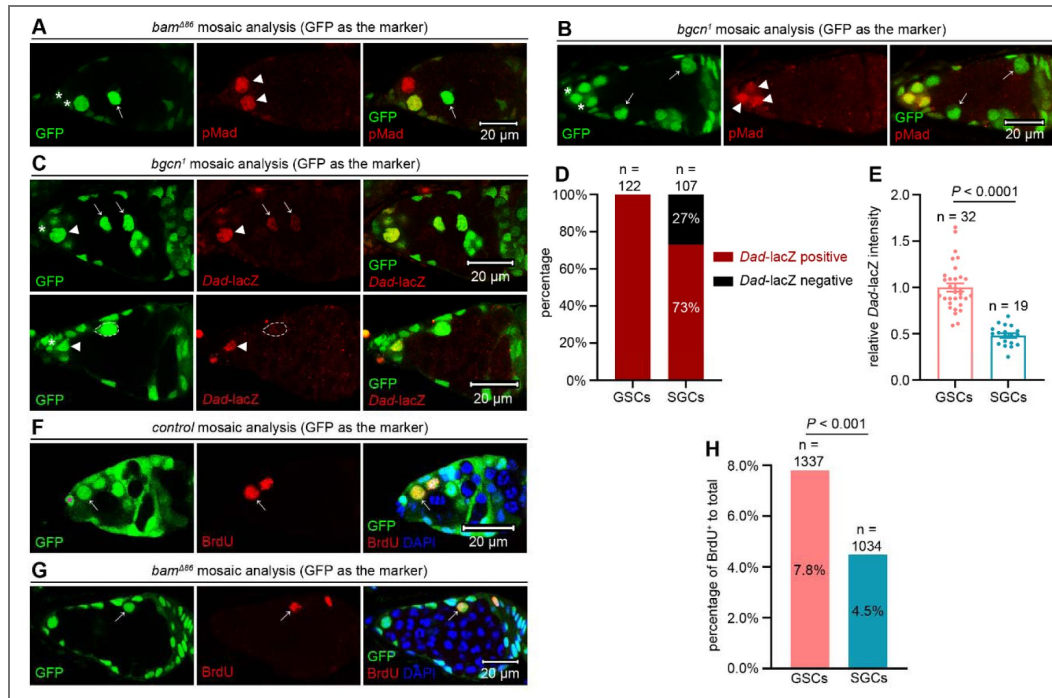


Figure 3. SGCs maintain lower BMP signaling levels than GSCs within the niche.

(A and B) Representative samples. The asterisks mark cap cells, arrowheads indicate pMad-positive GSCs, and arrows point to pMad-negative SGCs. (C) Representative samples. The asterisks denote cap cells, arrowheads mark *Dad-lacZ*-positive GSCs, and arrows highlight *Dad-lacZ*-positive SGCs. The dotted circles outline one *Dad-lacZ*-negative SGC. (D and E) Quantification data. 14-day-old flies were used for the analyses. In (E), data represent mean ± SEM, and statistical significance was determined by t test. (F) Representative sample. The asterisk marks a cap cell, while the arrows indicate a *BrdU*⁺ GSC within the niche. (G) Representative sample. The arrow indicates a *BrdU*⁺ SGC surrounded by germline tumors. (H) Quantification data. 14-day-old flies were used for the analyses. Statistical significance was determined by chi-squared test. The online version of this article includes the following source data for figure 3: Source data 1. All genotypes. Source data 2. Raw quantification data.



(A) Schematic of the experimental strategy for (B-F). Genotypes were unambiguously distinguished using a triple-color system (red, yellow, and green). (B-D) Representative samples. The dotted cycles mark an SGC, while the solid lines outline germline cysts containing differentiating cystocytes. (E and F) Quantification data. 14-day-old flies were used for the analyses. (G) Schematic of the experimental strategy for (H-J). (H and I) Representative samples. The dotted lines mark an SGC, while the solid lines outline a germline cyst containing differentiating cystocytes. (J) Quantification data. 14-day-old flies were used for the analyses. For each experiment, three independent replicates were performed, with over 100 SGCs and germline cysts quantified per replicate. Data represent mean $\pm$ SEM, and statistical significance was determined by t test. The online version of this article includes the following source data for figure 4: Source data 1. All genotypes. Source data 2. Raw quantification data.

cystocytes was normalized to the signal in wild-type TF and cap cells. Strikingly, *bam* mutant germline tumor cells exhibited significantly elevated expression of both *dpp* and *gbb* compared to wild-type cystocytes (Figure 5A-D [↗](#)).

To more sensitively assess Dpp and Gbb expression, we performed real-time quantitative PCR (RT-qPCR) analyses in *bam* or *bgn* mutant ovaries, comparing samples with and without germline-specific knockdown of *dpp* or *gbb*. Detection of reduced transcript levels in knockdown conditions would confirm active expression of these genes in the respective genetic backgrounds. Consistent with the essential roles of these two genes in fly viability, ubiquitous knockdown using *act-GAL4* with either *dpp-RNAi* or *gbb-RNAi* caused lethality, which results also validated the efficacy of these RNAi lines. Notably, germline-specific knockdown of *dpp* or *gbb* significantly reduced their transcript levels compared to *yellow* (*y*) or *white* (*w*) knockdown controls (Figure 5E, F [↗](#)). Collectively, these findings demonstrate that *bam* or *bgn* mutant germline tumors secrete the BMP ligands, albeit at lower levels than TF and cap cells.

Dpp and Gbb secreted by germline tumors are required to inhibit SGC differentiation

Finally, we investigated whether Dpp and Gbb secreted by germline tumors are required to inhibit SGC differentiation. Using a previously established double-mutant mosaic-analysis strategy for two genes on different chromosomes (Zhang et al., 2024 [↗](#); Zhang et al., 2023 [↗](#)), we generated *dpp bam* or *gbb bam* double-mutant germline clones using two *dpp* mutant alleles, *dpp^{d6}*, *dpp^{d12}*, and one *gbb* allele, *gbb¹* (Figure 6A, B [↗](#), see genotypes in Source data 1 [↗](#)). Heterozygotes in any of these alleles did not affect GSC maintenance, germ cell differentiation, and female fly fertility (Figure 6—figure supplement 1 [↗](#)). However, both *dpp bam* and *gbb bam* double-mutant germline tumor cells exhibited reduced proliferation rates compared to *bam* single-mutant controls (Figure 6—figure supplement 2 [↗](#)), indicating that autocrine BMP signaling promotes *bam* mutant tumor growth. As mentioned earlier, our evaluation focused on germ cells that have exited the niche and are surrounded by germline tumors to quantify the SGC phenotype. Thus, it raises the question of whether the extent of tumor encirclement (i.e., being surrounded by more or fewer tumor cells) influences the phenotype. To investigate this, we compared the SGC phenotype in bigger and smaller *bam* mutant germline tumors. Using confocal microscopy, we analyzed 70 germaria containing *bam* mutant germline clones that met our criteria for SGC phenotype quantification. For each *bam* mutant germline clone, we scanned and measured its maximal 2D cross-sectional area as a proxy for its 3D size. The 35 bigger and 35 smaller clones were categorized as ‘bigger’ and ‘smaller’ tumors, respectively. Strikingly, the SGC phenotype remained consistent between the two tumor groups (Figure 6—figure supplement 3 [↗](#)), aligning with our earlier finding that this phenotype is stable over a 14-day period (Figure 1J [↗](#)), a timeframe sufficient for substantial germline tumor growth. These results suggest that direct contact between tumorous and wild-type germ cells, rather than tumor size, is the primary determinant of this phenotype.

The results above demonstrate that comparing the severity of the SGC phenotype is feasible between germ cells surrounded by smaller *dpp bam* or *gbb bam* double-mutant germline tumors and those surrounded by larger *bam* single-mutant germline tumors. Remarkably, both *dpp bam* and *gbb bam* double-mutant germline tumors enclosed fewer SGCs but more germline cysts than their *bam* single-mutant counterparts (Figure 6C-I [↗](#)). Thus, we concluded that the BMP ligands from directly-contacting germline tumor cells mediate the dominant inhibition of SGC differentiation. This amazingly parallels the mechanism observed in normal stem cell niche, where only germ cells in direct contact with cap cells are maintained as GSCs (Chen and McKearin, 2003a [↗](#); Song et al., 2004 [↗](#); Xie and Spradling, 2000 [↗](#)).

Discussion

Our study reveals that *bam* or *bgn* mutant germline tumors in *Drosophila* ovaries secrete lower levels of BMP ligands Dpp and Gbb than cap cells, resulting in moderate BMP signaling activation in adjacent wild-type GSCs (called SGCs in this study). Such BMP signaling activation is sufficient to repress *bam* transcription, thereby blocking SGC differentiation (see our working model in Figure

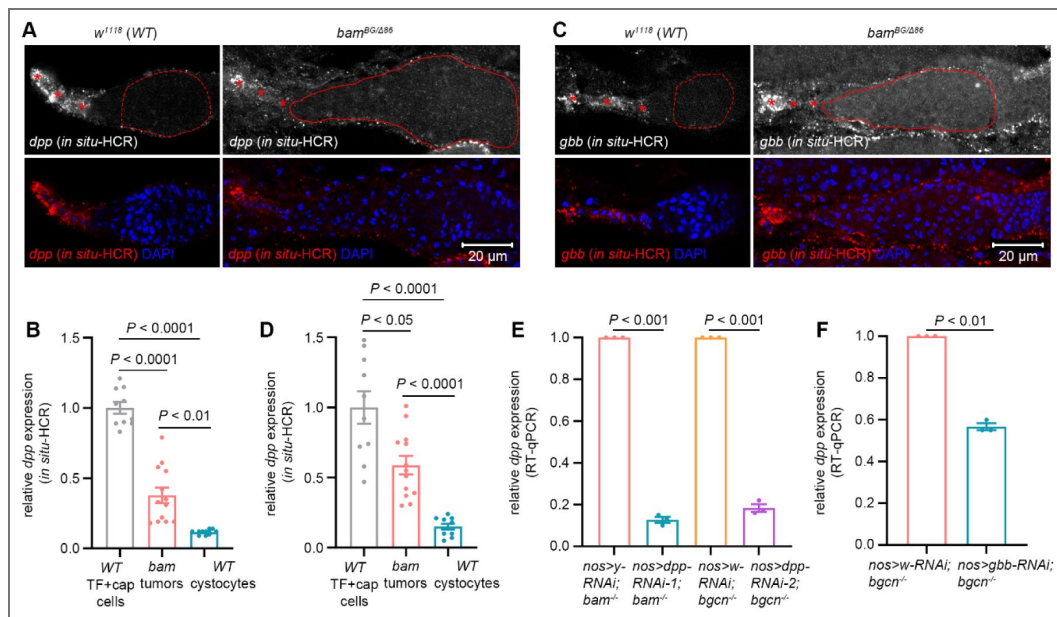


Figure 5. Germline tumors secrete Dpp and Gbb.

(A and C) Representative samples. The asterisks denote TF or cap cells. The dotted lines highlight *wild-type* (WT) cystocytes, while the solid lines outline *bam* mutant germline tumor cells. (B and D) Quantification data for *in situ*-HCR assays. 14-day-old flies were used for the analyses, and over 10 samples were quantified for each experiment. The *dpp* or *gbb* expression levels in both WT cystocytes and *bam* mutant germline tumor cells were normalized to the average expression levels in WT TF and cap cells. (E and F) Quantification data for RT-qPCR assays. 14-day-old flies were used for the analyses. For each experiment, three independent replicates were performed. Data represent mean $\pm$ SEM, and statistical significance in (B and D) was determined by one-way ANOVA and in (E and F) by t test. The online version of this article includes the following source data for figure 5 [Source data 1](#). [All genotypes](#). [Source data 2](#). [Raw quantification data](#).

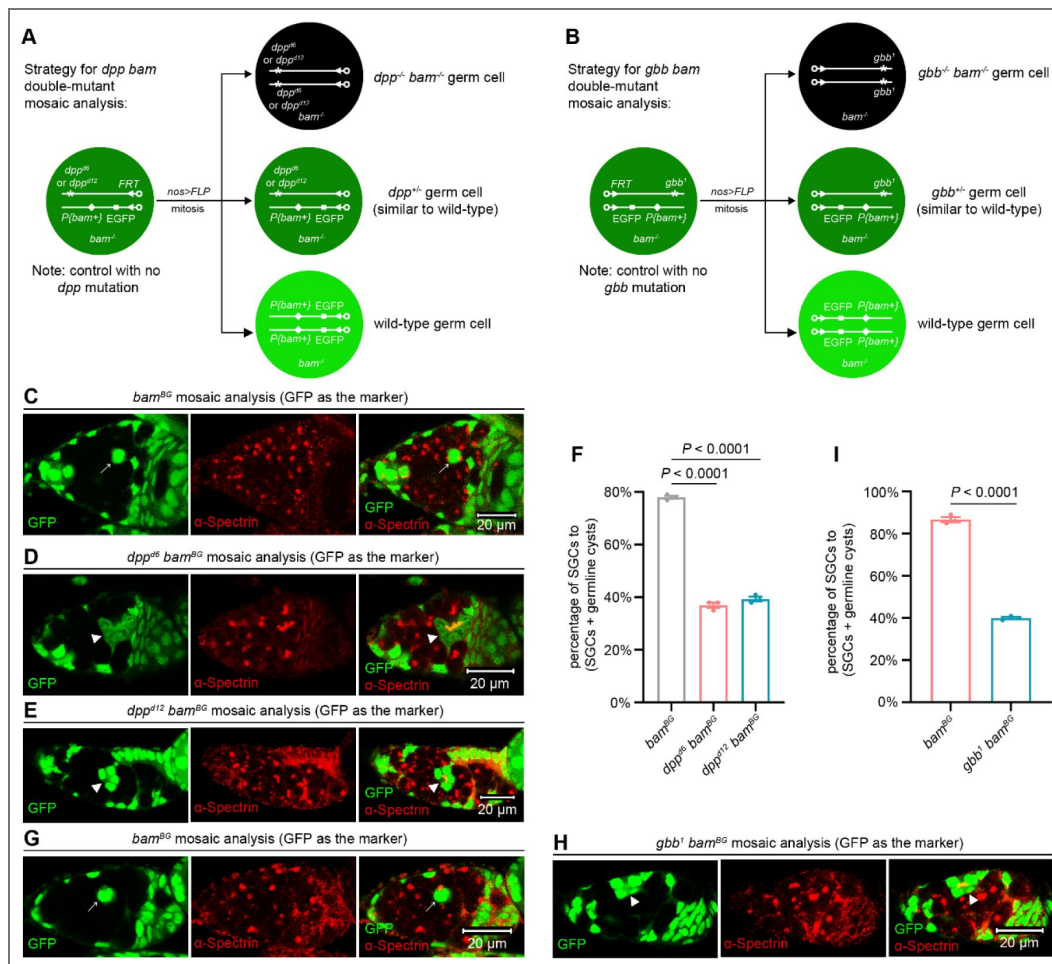


Figure 6. Dpp and Gbb secreted by germline tumors are required to inhibit SGC differentiation.

(A) Schematic of the experimental strategy for (C-F). (B) Schematic of the experimental strategy for (G-I). (C-E, G, and H) Representative samples. The arrows mark SGCs containing dot-like spectrosomes, while the arrowheads denote germline cysts with differentiating cystocytes that possess branched fusomes. (F and I) Quantification data for the SGC phenotype. 14-day-old flies were used for the analyses. For each experiment, three independent replicates were performed, with over 100 SGCs and germline cysts quantified per replicate. Data represent mean $\pm$ SEM. Statistical significance in (F) was determined by one-way ANOVA and in (I) by t test. The online version of this article includes the following source data and figure supplement for figure 6: [Source data 1](#). [All genotypes](#). [Source data 2](#). [Raw quantification data](#). [Figure supplement 1](#). [Monoallelic deletion of *dpp* or *gbb* does not affect GSC maintenance, germ cell differentiation, and female fly fertility](#). [Figure supplement 2](#). [dpp bam or gbb bam double-mutant germline tumor cells divide more slowly than bam single-mutant ones](#). [Figure supplement 3](#). [The SGC phenotype is unchanged irrespective of the number of surrounding germline tumors](#).

7). Strikingly, this mechanism closely recapitulates the normal niche signaling program mediated by TF and cap cells (Chen and McKearin, 2003a; Song et al., 2004; Xie and Spradling, 1998, 2000). To our knowledge, this represents the first evidence that tumor cells can functionally mimic a stem cell niche to arrest neighboring wild-type stem cells in an undifferentiated state.

While *bam* or *bgn* mutant germline tumors consist of GSC-like cells expected to resemble SGCs (Lavoie et al., 1999; McKearin and Ohlstein, 1995), we found key differences in BMP signaling. Out-of-niche *bgn* mutant tumor cells showed significantly lower BMP activity than neighboring SGCs, as evidenced by reduced *Dad-lacZ* expression (Figure 3C). Consistent with this, most of out-of-niche *bam* mutant tumor cells expressed *bamP*-GFP, a reporter suppressed by BMP signaling (Chen and McKearin, 2003a; Song et al., 2004), whereas only 26% of SGCs were positive (Figure 2B, C). These findings suggest that SGCs are more responsive to BMP signals secreted by germline tumors than the tumors themselves. Future studies are needed to elucidate the underlying mechanisms.

One interesting finding is that *bam* or *bgn* mutant germline tumors secrete lower levels of BMP ligands than TF and cap cells (Figure 5A-D). This aligns with earlier microarray data showing that purified *Drosophila* female GSCs express minimal *Dpp* and *Gbb* (Kai et al., 2005). However, our work reveals that such BMP levels in germline tumors are functionally critical to suppress SGC differentiation (Figure 6). Unlike normal GSCs, which receive unidirectional BMP ligands from cap cells (Chen and McKearin, 2003a; Li et al., 2016; Song et al., 2004; Xie and Spradling, 2000), SGCs are often fully surrounded by *bam* or *bgn* mutant germline tumors. This spatial advantage likely enables tumors to inhibit SGC differentiation efficiently without matching the high BMP output of TF and cap cells. Moreover, since BMP signaling is known to both inhibit normal GSC differentiation and promote their proliferation (Xie and Spradling, 1998), it should similarly stimulate SGC expansion, which is detrimental for *bam* or *bgn* mutant germline tumors. We propose that these tumor cells finely regulate BMP secretion to balance these opposing demands: maintaining differentiation blockade of SGCs while avoiding stimulation of their excessive proliferation.

A well-established principle in oncology is that tumor aggressiveness correlates with poor differentiation, with less-differentiated tumors exhibiting enhanced transformative capacity and metastatic potential (Jogi et al., 2012; Lytle et al., 2018). In *Drosophila* ovaries, *bam* or *bgn* mutant germline tumors consist of GSC-like cells that may resemble these poorly differentiated human tumors (Lavoie et al., 1999; McKearin and Ohlstein, 1995). This similarity raises the possibility that stem cell-like human tumors may similarly inhibit the differentiation of adjacent wild-type stem cells. By blocking differentiation, such tumors could deplete terminally differentiated cell populations, potentially exacerbating patient mortality. This mechanism may contribute to the heightened lethality of poorly differentiated tumors. Further investigation is needed to test this hypothesis.

The differentiation of a single GSC into a 16-cell germline cyst, comprising 15 polyploid nurse cells and one developing oocyte, represents a substantial metabolic investment (Fuller and Spradling, 2007; Lin, 1997). We propose that *bam* or *bgn* mutant germline tumors block this process to divert nutrients toward their own uncontrolled growth. This phenomenon could have broad implications, as many human tissues and organs (intestine, muscle, skin, blood system, male germline, etc.) similarly depend on adult stem cells for homeostasis (Blanpain and Fuchs, 2006; Gehart and Clevers, 2019; Sousa-Victor et al., 2022; Spradling et al., 2011; Wilkinson et al., 2020). Notably, these stem cell-dependent tissues and organs are frequent sites of tumorigenesis, raising the possibility that human cancers may similarly impair neighboring stem cell differentiation to optimize nutrient allocation for malignant growth. A key limitation of our study is that the evidence is derived solely from *Drosophila* germline. Future work should explore whether similar regulatory paradigms operate in mammalian tissues during tumorigenesis.

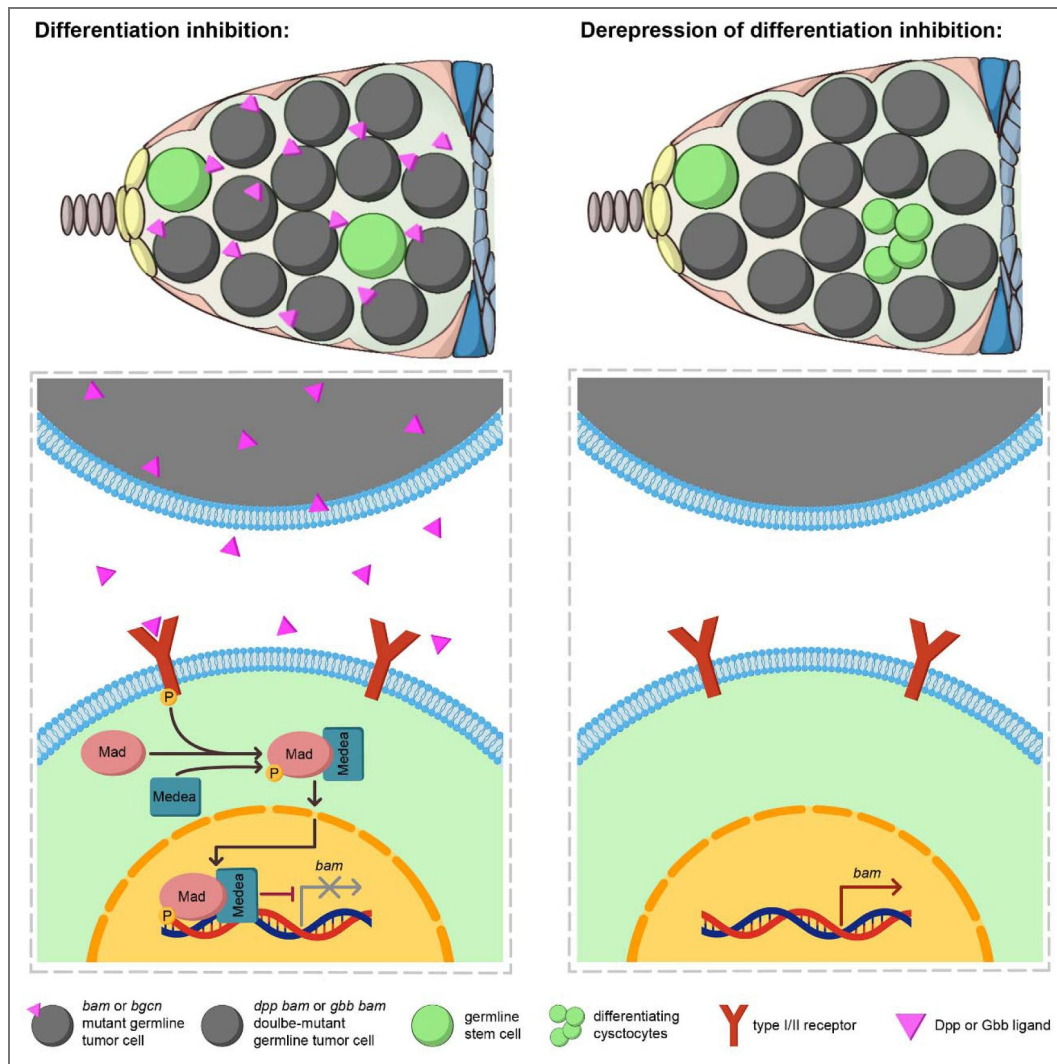


Figure 7. A working model.

bam or *bagn* mutant germline tumors secrete the BMP ligands Dpp and Gbb to activate BMP signaling in out-of-niche GSCs (called SGCs in this study) to inhibit their differentiation (left panel). In contrast, *dpp bam* and *gbb bam* double-mutant germline tumors exhibit a significant loss of this differentiation-inhibiting ability (right panel).

Materials and Methods

Fly husbandry

Flies were raised at 25°C on standard cornmeal/molasses/agar media.

Transgenic flies

hs-bam on chromosome 3R: The coding sequence of the *bam* gene, amplified from the cDNA clone, was cloned into the *BglII-XbaI* sites of the *pCaSpeR-hs* vector, while the *attB* sequence was inserted into the *XhoI* site. The resulting *attB-pCaSpeR-hs-bam* plasmid was then microinjected into the *attP154* (Chromosome 3R, 97D2) fly strain to generate site-specific transgenic flies.

Heatshock method to induce germline clones

To ensure developmental synchrony and maintain low-density growth, eggs within 8 hours of laying were collected for heatshock treatment. The animals (late-Larva 3/early-Pupa stage) were subjected to twice-daily heatshocks at 37°C (2 hours per session, with a 6-hour interval between the two sessions) for 6 consecutive days.

Fertility test

For each genotype, three independent crosses were performed. Each cross vial contained two females and four *w¹¹¹⁸* (wild-type) males, all aged three days old. The crosses were transferred to fresh vials every two days, with five replicate vials quantified per genotype. After all adult flies eclosed, offspring production was assessed by counting the number of empty pupae on the vial walls.

BrdU labeling

Ovaries were dissected in Schneider's insect medium (SIM) and incubated in freshly prepared BrdU solution (100 µg/mL in SIM) for five hours at 25°C. After washing with PBS for 30 min, samples were fixed in 4% paraformaldehyde (in PBS) for three hours, followed by another PBS wash for 30 min. Samples were then treated with RQ1 DNase reaction solution (Promega, Madison, WI, USA) for one hour, washed with PBST (0.3% Triton X-100 in PBS) for 30 min, and incubated overnight at 4°C with mouse anti-BrdU antibody. Following a PBST wash for one hour, ovaries were incubated with goat anti-mouse 546 and DAPI (0.1 µg/mL) in PBST for three hours, washed again in PBST for one hour, and mounted in autoclaved 70% glycerol.

Immuno-florescent staining, image collection, and data Processing

Ovaries were dissected in PBS, fixed in 4% paraformaldehyde (in PBS) for three hours, washed with PBST for 30 min, and then incubated overnight at 4°C with primary antibodies diluted in PBST as follows: mouse anti-BamC (1:5), mouse anti-β-Gal (1:200), mouse anti-α-Spectrin (1:100), rabbit anti-pMad (1:500), and rabbit anti-Vasa (1:2000). After washing with PBST for one hour, samples were incubated with Alexa Fluor-conjugated secondary antibodies (1:1000) and 0.1 µg/mL DAPI (in PBST) for three hours, followed by a final PBST wash for one hour. Ovaries were mounted in autoclaved 70% glycerol and imaged using a Zeiss LSM 710 confocal microscope (Carl Zeiss AG, Baden-Württemberg, Germany). Images were processed with ZEN 3.0 SR imaging software (Carl Zeiss) and Adobe Photoshop 2022. The quantification data were processed by GraphPad Prism, ImageJ, or Microsoft Excel.

In situ hybridization chain reaction (HCR)

Ovaries dissected from 14-day-old female flies were processed according to the following protocol.

1. Fixation: Ovaries were fixed in 4% paraformaldehyde (in PBS) for 3 hours at room temperature (RT) or overnight at 4°C;

Reagent or Resource	Source	Identifier
Antibodies		
mouse anti- α -Spectrin	Developmental Studies Hybridoma Bank (DSHB)	3A9
mouse anti-BamC	DSHB	AB_10570327
mouse anti- β -Gal	DSHB	JIE7
mouse anti-BrdU	Sigma	B5002
rabbit anti-pMad	Ed Laufer	N/A
rabbit anti-Vasa	Zhaohui Wang	(<i>Chen et al., 2014</i>)
goat anti-mouse 546	Invitrogen	A11003
goat anti-rabbit 488	Apexbio	K1206
goat anti-rabbit 546	Invitrogen	A11035
cDNA clone		
<i>bam</i> cDNA clone	Berkeley <i>Drosophila</i> Genome Project (BDGP)	FI05606
<i>Drosophila</i> strains		
<i>act-GAL4</i>	Zhaohui Wang	N/A
<i>bam</i> ^{BG} (strong loss-of-function allele)	Zhaohui Wang	(<i>Chen and McKearin, 2005</i>)
<i>bam</i> ^{A86} (null allele)	Michael Buszczak	(<i>Bopp et al., 1993</i>)

[Key resources table.](#)

<i>bamP-GFP</i>	Ting Xie	(Chen and McKearin, 2003b)
<i>bgn¹</i> (null allele)	Bloomington <i>Drosophila</i> Stock Center (BDSC)	6054
<i>bgn^{M06696}</i> (strong loss-of-function allele)	BDSC	40815
<i>Canton-S</i>	BDSC	64349
<i>Dad-lacZ</i>	Zheng Guo	(Kai and Spradling, 2003)
<i>dpp^{d6}</i> (hypomorphic allele)	BDSC	2062
<i>dpp^{d12}</i> (hypomorphic allele)	BDSC	2070
<i>dpp(P4)-lacZ</i>	Rongwen Xi	(Li et al., 2016)
<i>EGFP FRT40A</i>	BDSC	5629
<i>FRT40A</i>	BDSC	1816
<i>FRT42D</i>	BDSC	1802
<i>FRT42D EGFP</i>	BDSC	5626
<i>FRT82B</i>	BDSC	86313
<i>FRT82B arm-lacZ</i>	BDSC	7369
<i>FRT82B EGFP</i>	BDSC	32655
<i>FRT82B RFP</i>	BDSC	30555
<i>gbb¹</i> (null allele)	BDSC	98344
<i>hs-bam on chromosome 3R</i>	this study	N/A
<i>mad¹²</i> (null allele)	BDSC	51301
<i>med¹</i> (null allele)	BDSC	9033
<i>nos-GAL4-VP16</i>	Ruth Lehmann	(Van Doren et al., 1998)
<i>P{bam+}</i>	Shaowei Zhao	(Zhang et al., 2023)

Key resources table. (continued)

<i>punt</i> ¹³⁵ (strong loss-of-function allele)	BDSC	3100
<i>UASp-dpp-RNAi-1</i>	TsingHua Fly Center (THFC)	TH201500984.S
<i>UASp-dpp-RNAi-2</i>	THFC	THU5880
<i>UASp-FLP</i>	Shaowei Zhao	(<i>Zhang et al., 2023</i>)
<i>UASp-gbb-RNAi</i>	THFC	THU1480
<i>UASp-GFP</i>	Michael Buszczak	N/A
<i>UASp-yellow-RNAi</i>	THFC	TH03150.N
<i>UASz-FLP</i>	Shaowei Zhao	(<i>Zhang et al., 2023</i>)
<i>w</i> ¹¹¹⁸	Zhaohui Wang	N/A
Kits		
ChamQ SYBR qPCR Master Mix	Vazyme	Q311
HiFiScript cDNA Synthesis Kit	CWBIO	CW2569M
RNeasy Micro Kit	Qiagen	74004
Plasmids		
<i>pCaSpeR-hs</i>	Eric Baehrecke	N/A
<i>attB-pCaSpeR-hs-bam</i>	this study	N/A
Software		
Adobe Photoshop 2022	San Jose, CA, USA	
ImageJ	NIH	
Prism	Graphpad Software, Inc.	
ZEN 3.0 SR imaging software	Carl Zeiss	

Key resources table. (continued)

2. Hybridization: Following fixation, samples were washed three times for 5 min each in PBST, dehydrated in methanol for 5 min, rehydrated through a methanol:PBST gradient series (3:1, 1:1, and 1:3), followed by another three 5-min PBST washes, treated with Proteinase K (10 µg/mL) for 5 min, washed again three times for 5 min each in PBST, pre-hybridized in preheated hybridization buffer [50% formamide, 5× SSC, 9 mM citric acid (pH 6.0), 0.1% Tween 20, 50 µg/mL heparin, 1× Denhardt's solution, 10% dextran sulfate] for 30 min at 37°C, and then incubated with RNA-FISH probes (0.1 µM in hybridization buffer) overnight at 37°C.
3. Signal amplification: The next day, samples were washed four times for 15 min each at 37°C with preheated probe wash buffer (50% formamide, 5× SSC, 9 mM citric acid, 0.1% Tween 20, 50 µg/mL heparin), followed by three 10-min washes in 5× SSCT (5× SSC, 0.1% Tween 20) at RT. After pre-hybridization in amplification buffer (5× SSC, 0.1% Tween 20, 10% sodium sulfate) for 10 min at RT, an amplification reaction was performed using heat-denatured hairpin *h1/h2* nucleic acids (30 nM for each in amplification buffer) overnight in the dark at RT.
4. Washing and mounting: Samples were washed three times for 10 min each in 5× SSCT, followed by three 10-min washes in PBST, and then mounted in autoclaved 70% glycerol for imaging.

For the detection of *dpp* and *gbb*, a pool of 20 RNA-FISH probes targeting each mRNA was employed. The sequences of these probes were as follows:

dpp RNA-FISH probes:

GAGGAGGGCAGCAAACGGaaAGAGCATGGCCACGCTGTCCAGTTG
 GCACCACCGTACTTTGGTCGTTGAGtaGAAGAGTCTTCCTTTACG
 GAGGAGGGCAGCAAACGGaaTCGTAGCCCAGAGGCGCCACAATCC
 GGGCACTTCCCGTGGCAGTAATATGtaGAAGAGTCTTCCTTTACG
 GAGGAGGGCAGCAAACGGaaTCTCGGCTGCCGCTTGTTCGGCCG
 GTCGTGGTTCTTGCGCCTCGTAGGCTaGAAGAGTCTTCCTTTACG
 GAGGAGGGCAGCAAACGGaaGCTGCTTGCTGCCACGCTCGTG
 CGTCGTCCGTGTAGGTGAACAGGAGtaGAAGAGTCTTCCTTTACG
 GAGGAGGGCAGCAAACGGaaGGCCGGCTGGACATCGAGGCTCACC
 CTGCGGACTCGCCAGCCACCGGTCCtaGAAGAGTCTTCCTTTACG
 GAGGAGGGCAGCAAACGGaaCACCTGGTAGCGCGTCCGATTGCGC
 CCCGACGCGCGTGATGTCGTAGACtaGAAGAGTCTTCCTTTACG
 GAGGAGGGCAGCAAACGGaaTGCTCTTCACGTCAAGTGCAGCCG
 CCGCCTTCAGCTTCTCGTCGGCGGGtaGAAGAGTCTTCCTTTACG
 GAGGAGGGCAGCAAACGGaaCCCATGATCTCGGCGTAGAGCTTCT
 GGGATGTTGACCGAGTCGAGCTCGTtaGAAGAGTCTTCCTTTACG
 GAGGAGGGCAGCAAACGGaaAGCGCTCCTTGCTGTAGGTGGACG
 GGGTCTGGCTTCAGCTTGTCTTGAtaGAAGAGTCTTCCTTTACG
 GAGGAGGGCAGCAAACGGaaCACGAAGATTGATTCAATCGACGAG
 GCGGTCGAGCACCAGCGTCGGCTCCtaGAAGAGTCTTCCTTTACG

gbb RNA-FISH probes:

GAGGAGGGCAGCAAACGGaaGGTGGTACAGAACGGGTAGTGCTCC
 TTTTCAGGTTACATTCTCGTCGTTtaGAAGAGTCTTCCTTTACG
 GAGGAGGGCAGCAAACGGaaGCGTTCATGTGCGCATGAGCGGGA
 AGGGTCTGGACGATCGCATGGTTCGtaGAAGAGTCTTCCTTTACG

GAGGAGGGCAGCAAACGGaaGTACAGGGTCTGCATCTGGCAGCTG
ATGCCAGCCAGATCCTTGAAGTCTtaGAAGAGTCTTCCTTTACG
GAGGAGGGCAGCAAACGGaaTGCTCCTGTGGTGGCTGCTGTGGGC
GCTTGCGTGGATGGCTGGCGCTTCGtaGAAGAGTCTTCCTTTACG
GAGGAGGGCAGCAAACGGaaATGTCGTCCAGCTTCACCTCGCGGT
TCGTCCACCTTGGCGTGGATCAGTCtaGAAGAGTCTTCCTTTACG
GAGGAGGGCAGCAAACGGaaGTTGAGCTCCAACCAGCCCACGTAG
CAGCCACTCGTGCAGGCCCTCGGTtaGAAGAGTCTTCCTTTACG
GAGGAGGGCAGCAAACGGaaACTCCTGTTGGCGGTACGCCACTT
TGCCAATGGCGTATACCGTGATGGTtaGAAGAGTCTTCCTTTACG
GAGGAGGGCAGCAAACGGaaCGACGGCCGTGCTCGTGACGCAGTT
GGCACGTTGGAGACGTCAACCACataGAAGAGTCTTCCTTTACG
GAGGAGGGCAGCAAACGGaaGTTCTTCTGCTGCTCGCCCTCATCC
GGCCCGCTTGTCCAGGTCGGTGATGtaGAAGAGTCTTCCTTTACG
GAGGAGGGCAGCAAACGGaaTGATGCGGTGGTAGACGTCCAGCAG CCTGATCGCTGAGACCCTCCTCCGct
aGAAGAGTCTTCCTTTACG

The hairpin *h1/h2* sequences were as follows:

B1H1-594:

CGTAAAGGAAGACTCTTCCGTTTGCTGCCCTCCTCGCATTCTTTCTTGAGGAGGG
CAGCAAACGGGAAGAG

B1H2-594:

GAGGAGGGCAGCAAACGGGAAGAGTCTTCCTTTACGCTCTTCCGTTTGCTGCCCT
CCTCAAGAAAGAATGC

Real-time quantitative PCR (RT-qPCR)

Ovaries from 14-day-old flies were dissected, and total RNA was extracted using the RNeasy Micro Kit. Equal amounts of RNA were reverse-transcribed into cDNA using the HiFiScript cDNA Synthesis Kit. RT-qPCR was performed on a CFX Connect Real-Time PCR System (Bio-Rad) with ChamQ SYBR qPCR Master Mix. The PCR protocol consisted of an initial denaturation at 95°C for 30 min, followed by 40 cycles of 95°C for 10 sec and 60°C for 30 sec. Relative gene expression was calculated using the $2^{-\Delta\Delta CT}$ method (Livak and Schmittgen, 2001 [↗](#)). The primers described previously were used (Huang et al., 2017 [↗](#)):

dpp primer-1: TACCACGCCATCCACTCAAC

dpp primer-2: GCTCGTTACTCGATACGGCT

gbb primer-1: CTGGATCATCGCACCAGAGG

gbb primer-2: GTCTGGACGATCGCATGGTT

rp49 (internal control) primer-1: CACCGGATTCAAGAAGTTCC

rp49 (internal control) primer-2: GACAATCTCCTTGCGCTTCT

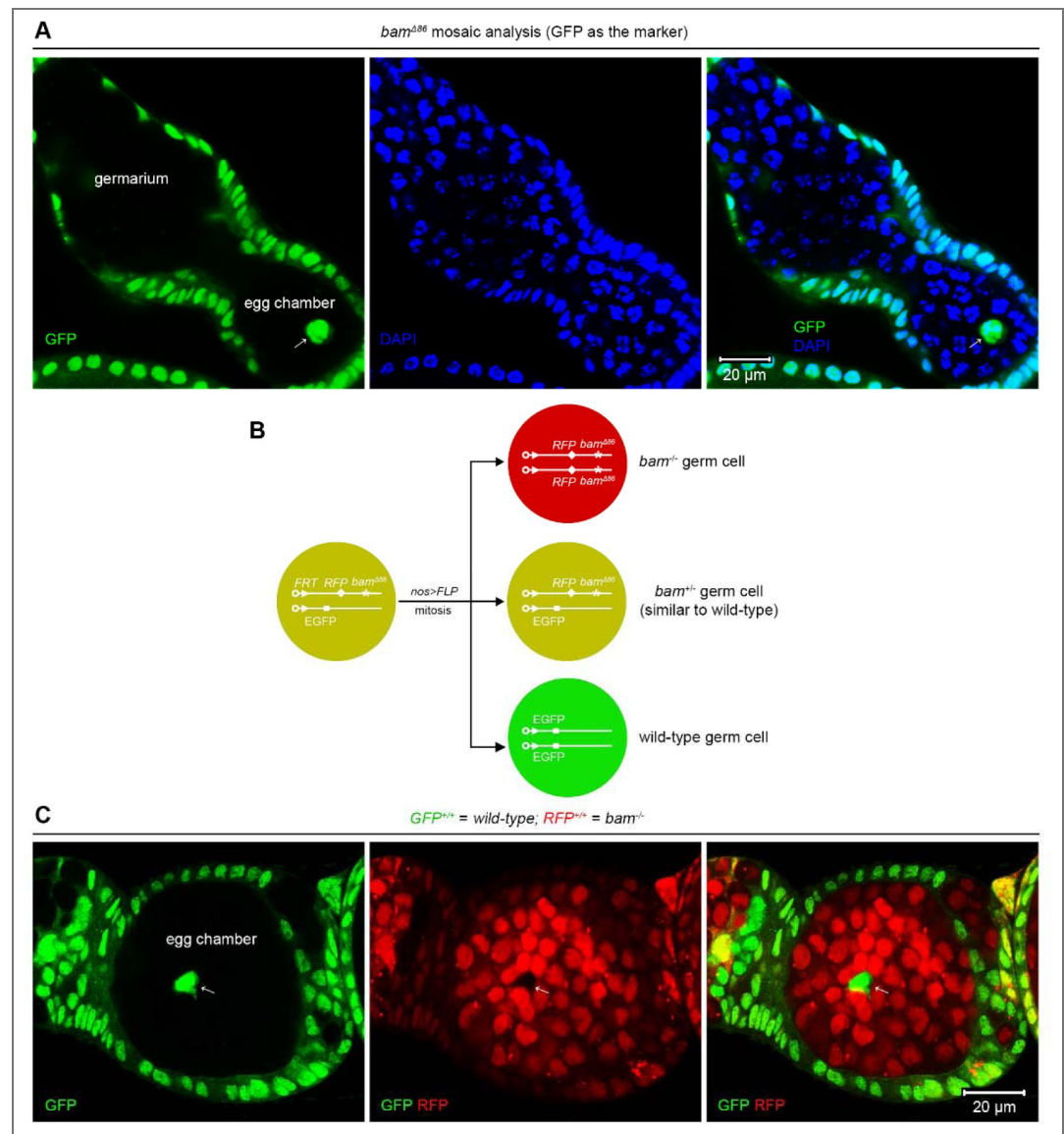


Figure 1-figure supplement 1. SGCs appear in egg chambers. (A, C) Representative samples. The arrows indicate SGCs enclosed within egg chambers. (B) Schematic of the experimental strategy for (C). See also [Source data 1](#).

Figure 6-figure supplement 1. Monoallelic deletion of *dpp* or *gbb* does not affect GSC maintenance, germ cell differentiation, and female fly fertility.

(A-D) Representative samples. (E) Quantification data. 14-day-old flies were used for the analyses. GSCs are germ cells that are located within the niche and contain dot-like spectrosomes. Cystoblasts are germ cells that have exited the niche, remain in contact with GSCs, and maintain dot-like spectrosomes. Cystocytes in germline cysts are germ cells that are characterized by branched fusomes. (F) Fertility test. 3-day-old flies were used for the analyses. For each genotype, three independent replicates were performed. Data represent mean $\pm$ SEM, and statistical significance was determined by one-way ANOVA. *n.s.* ($P > 0.05$). See also Source data 1 and 2.

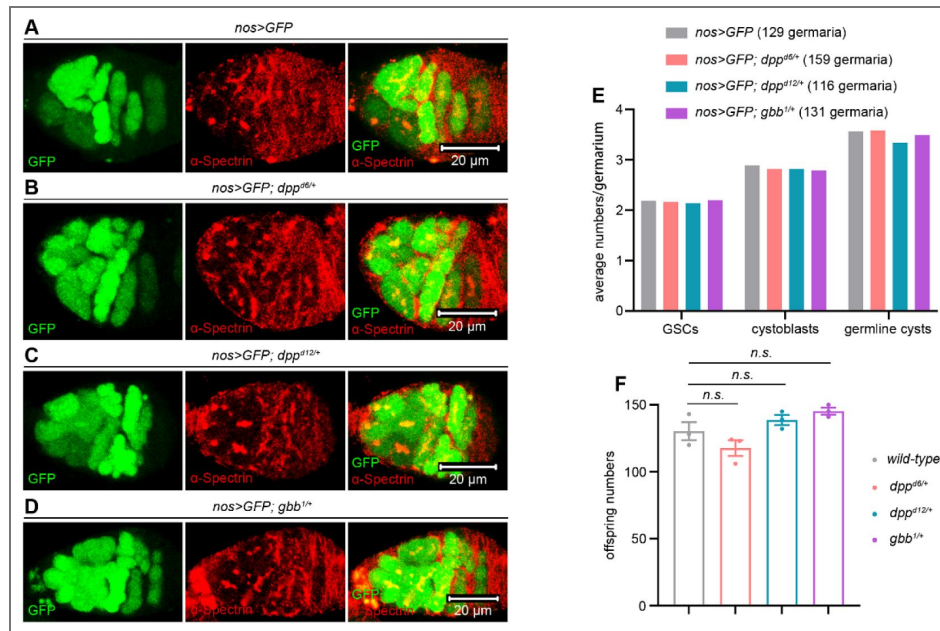
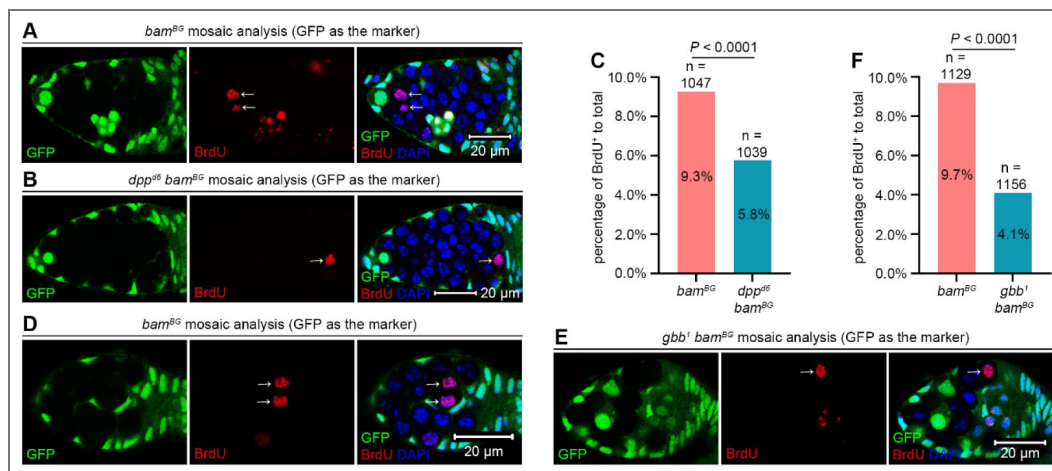


Figure 6-figure supplement 2. *dpp bam* or *gbb bam* double-mutant germline tumor cells divide more slowly than *bam* single-mutant ones.

(A, B, D, and E) Representative samples. The arrows indicate BrdU⁺ germline tumor cells mutant for *bam*, *dpp bam*, or *gbb bam*. (C and F) Quantification data. 14-day-old flies were used for the analyses. Statistical significance was determined by chi-squared test. See also Source data 1 and 2.



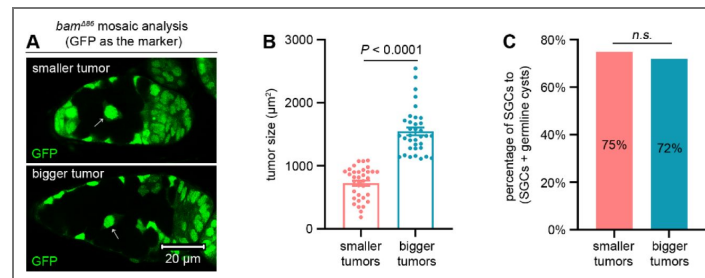


Figure 6-figure supplement 3. The SGC phenotype is unchanged irrespective of the number of surrounding germline tumors.

(A) Representative samples. The arrows denote SGCs. Both images are of the same magnification. (B) Quantification data for tumor size. Data represent mean $\pm$ SEM. (C) Quantification data for the SGC phenotype. In (B and C), 14-day-old flies were used for the analyses. Statistical significance in (B) was determined by t test and in (C) by chi-squared test. *n.s.* ($P > 0.05$). See also Source data 1 [1](#) and 2 [2](#).

Data availability

All genotypes are described in [Source data 1](#), and the raw quantification data are included in [Source data 2](#). Fly strains and plasmids are available upon request.

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Additional information

Author contributions

S.Z. conceived and supervised this study. Y.Z., Y.W., J.S., L.Y., Z.W., D.S., J.W., S.Y., L.N., C.S., H.Z., Y.D.Z., and S.Z. performed the experiments. S.Z. wrote the manuscript and all authors commented on it.

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Additional files

[Source data 1](#). All genotypes.

[Source data 2](#). Raw quantification data.

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Peer reviews

Reviewer #1 (Public review):

Summary:

This preprint from Shaowei Zhao and colleagues presents results that suggest tumorous germline stem cells (GSCs) in the *Drosophila* ovary mimic the ovarian stem cell niche and inhibit the differentiation of neighboring non-mutant GSC-like cells. The authors use FRT-mediated clonal analysis driven by a germline-specific gene (*nos-Gal4*, *UASp-flp*) to induce GSC-like cells mutant for *bam* or *bam*'s co-factor *bgn*. *Bam*-mutant or *bgn*-mutant germ cells produce tumors in the stem cell compartment (the germarium) of the ovary (Fig. 1). These tumors contain non-mutant cells - termed SGC for single-germ cells. 75% of SGCs do not exhibit signs of differentiation (as assessed by *bamP-GFP*) (Fig. 2). The authors demonstrate that block in differentiation in SGC is a result of suppression of *bam* expression (Fig. 2). They present data suggesting that in 73% of SGCs BMP signaling is low (assessed by *dad-lacZ*) (Fig. 3) and proliferation is less in SGCs vs GSCs. They present genetic evidence that mutations in BMP pathway receptors and transcription factors suppress some of the non-autonomous effects exhibited by SGCs within *bam*-mutant tumors (Fig. 4). They show data that *bam*-mutant cells secrete Dpp, but this data is not compelling (see below) (Fig. 5). They provide genetic data that loss of BMP ligands (*dpp* and *gbb*) suppresses the appearance of SGCs in *bam*-mutant tumors (Fig. 6). Taken together, their data support a model in which *bam*-mutant GSC-like cells produce BMPs that act on non-mutant cells (i.e., SGCs) to prevent their differentiation, similar to what is seen in the ovarian stem cell niche. This preprint from Shaowei Zhao and colleagues presents results that suggest tumorous germline stem cells

(GSCs) in the *Drosophila* ovary mimic the ovarian stem cell niche and inhibit the differentiation of neighboring non-mutant GSC-like cells. The authors use FRT-mediated clonal analysis driven by a germline-specific gene (*nos-Gal4*, *UASp-flp*) to induce GSC-like cells mutant for *bam* or *bam*'s co-factor *bgn*. *Bam*-mutant or *bgn*-mutant germ cells produce tumors in the stem cell compartment (the germarium) of the ovary (Fig. 1). These tumors contain non-mutant cells - termed SGC for single-germ cells. 75% of SGCs do not exhibit signs of differentiation (as assessed by *bamP-GFP*) (Fig. 2). The authors demonstrate that block in differentiation in SGC is a result of suppression of *bam* expression (Fig. 2). They present data suggesting that in 73% of SGCs BMP signaling is low (assessed by *dad-lacZ*) (Fig. 3) and proliferation is less in SGCs vs GSCs. They present genetic evidence that mutations in BMP pathway receptors and transcription factors suppress some of the non-autonomous effects exhibited by SGCs within *bam*-mutant tumors (Fig. 4). They show data that *bam*-mutant cells secrete Dpp, but this data is not compelling (see below) (Fig. 5). They provide genetic data that loss of BMP ligands (*dpp* and *gbb*) suppresses the appearance of SGCs in *bam*-mutant tumors (Fig. 6). Taken together, their data support a model in which *bam*-mutant GSC-like cells produce BMPs that act on non-mutant cells (i.e., SGCs) to prevent their differentiation, similar to what is seen in the ovarian stem cell niche.

Strengths:

- (1) Use of an excellent and established model for tumorous cells in a stem cell microenvironment
- (2) Powerful genetics allow them to test various factors in the tumorous vs non-tumorous cells
- (3) Appropriate use of quantification and statistics

Weaknesses:

- (1) What is the frequency of SGCs in *nos>flp*; *bam*-mutant tumors? For example, are they seen in every germarium, or in some germaria, etc or in a few germaria.

This concern was addressed in the rebuttal. The line number is 106, not line 103.

- (2) Does the breakdown in clonality vary when they induce *hs-flp* clones in adults as opposed to in larvae/pupae?

This concern was addressed in the rebuttal. However, these statements are not on lines 331-335 but instead starting on line 339. Please be accurate about the line numbers cited in the rebuttal. They need to match the line numbers in the revised manuscript.

- (3) Approximately 20-25% of SGCs are *bam*⁺, *dad-LacZ*⁺. Firstly, how do the authors explain this? Secondly, of the 70-75% of SGCs that have no/low BMP signaling, the authors should perform additional characterization using markers that are expressed in GSCs (i.e., *Sex lethal* and *nanos*).

The authors did not perform additional staining for GSC-enriched protein like *Sex lethal* and *nanos*.

- (4) All experiments except Fig. 1I (where a single germarium with no quantification) were performed with *nos-Gal4*, *UASp-flp*. Have the authors performed any of the phenotypic characterizations (i.e., figures other than figure 1) with *hs-flp*?

In the rebuttal, the authors stated that they used *nos>flp* for all figures except for Fig. 1I. It would be more convincing for them to prove in Fig. 1 that there is no phenotypic difference between the two methods and then switch to the *nos>FLP* method for the rest of the paper.

(5) Does the number of SGCs change with the age of the female? The experiments were all performed in 14-day old adult females. What happens when they look at young female (like 2-day old). I assume that the *nos>flp* is working in larval and pupal stages and so the phenotype should be present in young females. Why did the authors choose this later age? For example, is the phenotype more robust in older females? or do you see more SGCs at later time points?

The authors did not supply any data to prove that the clones were larger in 14-day-old flies than in younger flies. Additionally, the age of "younger" flies was not specified. Therefore, the authors did not satisfactorily answer my concern.

(6) Can the authors distinguish one copy of GFP versus 2 copies of GFP in germ cells of the ovary? This is not possible in the *Drosophila* testis. I ask because this could impact on the clonal analyses diagrammed in Fig. 4A and 4G and in 6A and B. Additionally, in most of the figures, the GFP is saturated so it is not possible to discern one vs two copies of GFP.

In the rebuttal, the authors stated that they cannot differential one vs two copies of GFP. They used other clone labeling methods in Fig. 4 and 6. I think that the authors should make a statement in the manuscript that they cannot distinguish one vs two copies of GFP for the record.

(7) More evidence is needed to support the claim of elevated Dpp levels in *bam* or *bgn* mutant tumors. The current results with *dpp-lacZ* enhancer trap in Fig 5A,B are not convincing. First, why is the *dpp-lacZ* so much brighter in the mosaic analysis (A) than in the no-clone analysis (B); it is expected that the level of *dpp-lacZ* in cap cells should be invariant between ovaries and yet LacZ is very faint in Fig. 5B. I think that if the settings in A matched those in B, the apparent expression of *dpp-lacZ* in the tumor would be much lower and likely not statistically significantly. Second, they should use RNA in situ hybridization with a sensitive technique like hybridization chain reactions (HCR) - an approach that has worked well in numerous *Drosophila* tissues including the ovary.

The HCR FISH in Fig.5 of the revised manuscript needs an explanation for how the mRNA puncta were quantified. Currently, there is no information in the methods. What is meant but relative *dpp* levels. I think that the authors should report in an unbiased manner "number" of *dpp* or *gbb* puncta in TFs. For the germaria, I think that they should report the number of puncta of *dpp* or *gbb* divide by the total area in square pixels counted. Additionally, the background fluorescence is noticeably much higher in *bamBG/delta86* germaria, which would (falsely) increase the relative intensity of *dpp* and *gbb* in *bam* mutants. Although, I commend the authors for performing HCR FISH, these data are still not convincing to me.

(8) In Fig 6, the authors report results obtained with the *bamBG* allele. Do they obtain similar data with another *bam* allele (i.e., *bamdelta86*)?

The authors did not try any experiments with the *bamdelta86* allele, despite this allele being molecularly defined, where the *bamBG* allele is not defined.

<https://doi.org/10.7554/eLife.108910.2.sa3>

Reviewer #2 (Public review):

In the current version, Zhang et al. have made substantial improvements to the manuscript. It is now easier to read, and the data are more solid compared with the previous version, supporting their conclusion that tumor GSCs secrete stemness factors (BMPs and Dpp) to suppress the differentiation of neighboring wild-type GSCs. This study should benefit a broad readership across developmental biology, germ cell biology, stem cell biology, and cancer biology.

However, the following suggestions may further improve the clarity and rigor of the research content:

(1) Clarification of sample size (n).

Each germarium can contain highly variable numbers of SGCs, sometimes reaching 50-100. When reporting "n" values, the authors are encouraged to also indicate the number of germaria analyzed. For example, in lines 126-128:

"Notably, 74% of SGCs (n = 132) were GFP-negative, while the remaining 26% were GFP-positive (Figure 2B, C). This suggests that SGCs can be categorized into two distinct groups: those resembling GSCs (GSC-like) and those resembling cystoblasts (cystoblast-like)."

Please clarify how many germaria were examined to obtain n = 132. In addition, it is unclear whether the authors intend to suggest that the GFP-negative SGCs are GSC-like or cystoblast-like; this point should be clarified.

(2) Improvement of Fig. 6 in situ hybridization images.

The in situ hybridization images in Fig. 6 are not fully convincing. The control images, in particular, would benefit from higher resolution and enlarged views of the germarium region. In panel C, abundant signals are also present outside the germarium, which may complicate interpretation and should be clarified or controlled for.

Alternatively, the authors could strengthen the in situ analysis by using bam mutants or bam dpp / bam gbb double mutants as controls to better define signal specificity.

<https://doi.org/10.7554/eLife.108910.2.sa2>

Reviewer #3 (Public review):

Zhang et al. investigated how germline tumors influence the development of neighboring wild-type (WT) germline stem cells (GSC) in the *Drosophila* ovary. They report that germline tumors generated by differentiation-arrested mutations (bam and bgcn) inhibit the differentiation of neighboring WT GSCs by arresting them in an undifferentiated state, resulting from reduced expression of the differentiation-promoting factor Bam. They find that these tumor cells produce low levels of the niche-associated signaling molecules Dpp and Gbb, which suppress bam expression and consequently inhibit the differentiation of neighboring WT GSCs non-cell-autonomously. Based on these findings, the authors propose that germline tumors mimic the niche to suppress the differentiation of the neighboring wild-type germline stem cells.

Strengths:

The study uses a well-established in vivo model to address an important biological question concerning the interaction between germline tumor cells and wild-type (WT) germline stem cells in the *Drosophila* ovary. If the findings are substantiated, this study could provide valuable insights that are applicable to other stem cell systems.

Weaknesses:

The authors have addressed some of my concerns in the revised submission. However, the data presented do not allow the authors to distinguish whether the failed differentiation of WT stem cells/germline cells results from "arrested differentiation due to the loss of the differentiation niche" or from "direct inhibition by tumor-derived expression of niche-associated molecules Dpp and Gbb". The critical supporting data, HCR in situ results, are not sufficiently convincing.

<https://doi.org/10.7554/eLife.108910.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife Assessment

This study presents results supporting a model that tumorous germline stem cells (GSCs) in the Drosophila ovary mimic the stem cell niche and inhibit the differentiation of neighboring cells. The valuable findings show that GSC tumors often contain non-mutant cells whose differentiation is suppressed by the GSC tumorous cells. However, the evidence showing that the GSC tumors produce BMP ligands to suppress differentiation of non-mutant cells is incomplete. It could be strengthened by the use of sensitive RNA in situ hybridization approaches.

Thank you for your valuable assessment. RNA in situ hybridization evidence has been added to the revised manuscript (Figure 5A-D) to support that GSC tumors produce BMP ligands.

Public Reviews:

Reviewer #1 (Public review):

Summary:

*This preprint from Shaowei Zhao and colleagues presents results that suggest tumorous germline stem cells (GSCs) in the Drosophila ovary mimic the ovarian stem cell niche and inhibit the differentiation of neighboring non-mutant GSC-like cells. The authors use FRT-mediated clonal analysis driven by a germline-specific gene (*nos-Gal4, UASp-flp*) to induce GSC-like cells mutant for *bam* or *bam*'s co-factor *bgn*. *Bam*-mutant or *bgn*-mutant germ cells produce tumors in the stem cell compartment (the germarium) of the ovary (Figure 1). These tumors contain non-mutant cells - termed SGC for single-germ cells. 75% of SGCs do not exhibit signs of differentiation (as assessed by *bamP-GFP*) (Figure 2). The authors demonstrate that block in differentiation in SGC is a result of suppression of *bam* expression (Figure 2). They present data suggesting that in 73% of SGCs, BMP signaling is low (assessed by *dad-lacZ*) (Figure 3) and proliferation is less in SGCs vs GSCs. They present genetic evidence that mutations in BMP pathway receptors and transcription factors suppress some of the non-autonomous effects exhibited by SGCs within *bam*-mutant tumors (Figure 4). They show data that *bam*-mutant cells secrete *Dpp*, but this data is not compelling (see below) (Figure 5). They provide genetic data that loss of BMP ligands (*dpp* and *gbb*) suppresses the appearance of SGCs in *bam*-mutant tumors (Figure 6). Taken together, their data support a model in which *bam*-mutant GSC-like cells produce BMPs that act on non-mutant cells (i.e., SGCs) to prevent their differentiation, similar to what is seen in the ovarian stem cell niche.*

Strengths:

- (1) Use of an excellent and established model for tumorous cells in a stem cell microenvironment.*
- (2) Powerful genetics allow them to test various factors in the tumorous vs non-tumorous cells.*
- (3) Appropriate use of quantification and statistics.*

We greatly appreciate your valuable comments.

Weaknesses:

(1) What is the frequency of SGCs in *nos>flp*; *bam*-mutant tumors? For example, are they seen in every germarium, or in some germaria, etc, or in a few germaria?

This is a good question. Because the SGC phenotype depends on the presence of both germline tumor clones and out-of-niche wild-type germ cells, our quantification was restricted to germaria containing both. In 14-day-old fly ovaries, 70% of germaria (432/618) met this criterion (Line 103). Each of them contained an average of 1.5 SGCs (Figure 1K).

(2) Does the breakdown in clonality vary when they induce *hs-flp* clones in adults as opposed to in larvae/pupae?

Our attempts to induce ovarian *hs-FLP* germline clones by heat-shocking adult flies were unsuccessful, with very few clones being observed. Therefore, we shifted our approach to an earlier developmental stage. Successful induction was achieved by subjecting late-L3/early-pupal animals to a twice-daily heatshock at 37°C for 6 consecutive days (2 hours per session with a 6-hour interval, see Lines 331-335) (Zhao et al., 2018).

(3) Approximately 20-25% of SGCs are *bam*⁺, *dad-lacZ*⁺. Firstly, how do the authors explain this? Secondly, of the 70-75% of SGCs that have no/low BMP signaling, the authors should perform additional characterization using markers that are expressed in GSCs (i.e., *Sex lethal* and *nanos*).

These 20-25% of SGCs are *bamP-GFP*⁺ *dad-lacZ*⁺, not *bam*⁺ *dad-lacZ*⁺ (see Figure 2C and 3D). They would be cystoblast-like cells that may have initiated a differentiation program toward forming germline cysts (see Lines 122-130). The 70-75% of SGCs that have low BMP signaling exhibit GSC-like properties, including: 1) dot-like spectrosomes; 2) *dad-lacZ* positivity; 3) absence of *bamP-GFP* expression. While additional markers would be beneficial, we think that this combination of properties is sufficient to classify these cells as GSC-like.

(4) All experiments except Figure 1I (where a single germarium with no quantification) were performed with *nos-Gal4*, *UASp-flp*. Have the authors performed any of the phenotypic characterizations (i.e., figures other than Figure 1) with *hs-flp*?

Yes, we initially identified the SGC phenotype through *hs-FLP*-mediated mosaic analysis of *bam* or *bgn* mutant in ovaries. However, as noted in our response to Weakness (2), this approach was very labor-intensive. Therefore, we switched to using the more convenient *nos>FLP* system for subsequent experiments. To our observation, there was no difference in inducing the SGC phenotype by these two approaches.

(5) Does the number of SGCs change with the age of the female? The experiments were all performed in 14-day-old adult females. What happens when they look at a young female (like 2-day-old). I assume that the *nos>flp* is working in larval and pupal stages, and so the phenotype should be present in young females. Why did the authors choose this later age? For example, is the phenotype more robust in older females? Or do you see more SGCs at later time points?

These are very good questions. The SGC phenotype was consistent over the 14-day analysis period (Figure 1J) and was specifically dependent on the presence of germline tumor clones. In 14-day-old fly ovaries, these clones were both larger and more frequent than in younger flies. This age-dependent enhancement in clone size and frequency significantly improved our quantification efficiency (see Lines 101-112).

(6) Can the authors distinguish one copy of GFP versus 2 copies of GFP in germ cells of the ovary? This is not possible in the *Drosophila* testis. I ask because this could impact the clonal analyses diagrammed in Figure 4A and 4G and in 6A and B. Additionally, in

most of the figures, the GFP is saturated, so it is not possible to discern one vs two copies of GFP.

Thank you for this valuable comment. It was also difficult for us to distinguish 1 and 2 copies of GFP in the *Drosophila* ovary. In Figure 4A-F, to resolve this problem, we used a triple-color system, in which red germ cells (RFP^{+/+} GFP^{-/-}) are *bam* mutant, yellow germ cells (RFP^{+/+} GFP^{+/+}) are wild-type, and green germ cells (RFP^{-/-} GFP^{+/+}) are *punt* or *med* mutant. In Figure 4G-J, we quantified the SGC phenotype only in black germ cells (GFP^{-/-}), which are wild-type (control) or *mad* mutant. In Figure 6, we quantified the SGC phenotype only in green germ cells (both GFP^{+/+} and GFP^{+/+}), all of which are wild-type.

(7) More evidence is needed to support the claim of elevated Dpp levels in bam or bgcn mutant tumors. The current results with the dpp-lacZ enhancer trap in Figure 5A, B are not convincing. First, why is the dpp-lacZ so much brighter in the mosaic analysis (A) than in the no-clone analysis (B)? It is expected that the level of dpp-lacZ in cap cells should be invariant between ovaries, and yet LacZ is very faint in Figure 5B. I think that if the settings in A matched those in B, the apparent expression of dpp-lacZ in the tumor would be much lower and likely not statistically significant. Second, they should use RNA in situ hybridization with a sensitive technique like hybridization chain reactions (HCR) - an approach that has worked well in numerous Drosophila tissues, including the ovary.

Thank you for this critical comment. The settings of immunofluorescent staining and confocal parameters in the original Figure 5A were the same as those in 5B. To our observation, the levels of *dpp-lacZ* in terminal filament and cap cells were highly variable across germaria, even within the same ovary. We have omitted these results from the revised Figure 5. Instead, the HCR-FISH data have been added (Figure 5A-D) to support that *bam* mutant germline tumors secrete BMP ligands.

(8) In Figure 6, the authors report results obtained with the bamBG allele. Do they obtain similar data with another bam allele (i.e., bamdelta86)?

No. Given that *bam*^{BG} was functionally indistinguishable from *bam*^{Δ86} in inducing the SGC phenotype (Figure 1J), we believe that repeating these experiments with *bam*^{Δ86} would be redundant and would not alter the key conclusion of our study. Thank you for your understanding!

Reviewer #2 (Public review):

While the study by Zhang et al. provides valuable insights into how germline tumors can non-autonomously suppress the differentiation of neighboring wild-type germline stem cells (GSCs), several conceptual and technical issues limit the strength of the conclusions.

Major points:

(1) Naming of SGCs is confusing. In line 68, the authors state that "many wild-type germ cells located outside the niche retained a GSC-like single-germ-cell (SGC) morphology." However, bam or bgcn mutant GSCs are also referred to as "SGCs," which creates confusion when reading the text and interpreting the figures. The authors should clarify the terminology used to distinguish between wild-type SGCs and tumor (bam/bgcn mutant) SGCs, and apply consistent naming throughout the manuscript and figure legends.

We apologize for any confusion. In our manuscript, the term "SGC" is reserved specifically for wild-type germ cells that maintain a GSC-like morphology outside the niche. *bam* or *bgcn* mutant germ cells are referred to as GSC-like tumor cells (Lines 89-90), not SGCs.

(a) The same confusion appears in Figure 2. It is unclear whether the analyzed SGCs are wild-type or bam mutant cells. If the SGCs analyzed are Bam mutants, then the lack of Bam expression and failure to differentiate would be expected and not informative. However, if the SGCs are wild-type GSCs located outside the niche, then the observation would suggest that Bam expression is silenced in these wild-type cells, which is a significant finding. The authors should clarify the genotype of the SGCs analyzed in Figure 2C, as this information is not currently provided.

The SGCs analyzed in Figure 2A-C are wild-type, GSC-like cells located outside the niche. They were generated using the same genetic strategy depicted in Figures 1C and 1E (with the schematic in Figure 1B). The complete genotypes for all experiments are available in Source data 1.

(b) In Figures 4B and 4E, the analysis of SGC composition is confusing. In the control germaria (bam mutant mosaic), the authors label GFP⁺ SGCs as "wild-type," which makes interpretation unclear. Note, this is completely different from their earlier definition shown in line 68.

The strategy to generate SGCs in Figure 4B-F (with the schematic in Figure 4A) is different from that in Figure 1C-F, H, and I (with the schematic in Figure 1B). In Figure 4B-F, we needed to distinguish *punt*^{-/-} (or *med*^{-/-}) with *punt*^{+/-} (or *med*^{+/-}) germ cells. As noted in our response to Reviewer #1's Weakness (6), it was difficult for us to distinguish 1 and 2 copies of GFP in the Drosophila ovary. Therefore, we chose to use the triple-color system to distinguish these germ cells in Figure 4B-F (see genotypes in Source data 1).

(c) Additionally, *bam*^{+/-} GSCs (the first bar in Figure 4E) should appear GFP⁺ and Red^{>sup>+} (i.e., yellow). It would be helpful if the authors could indicate these *bam*^{+/-} germ cells directly in the image and clarify the corresponding color representation in the main text. In Figure 2A, although a color code is shown, the legend does not explain it clearly, nor does it specify the identity of *bam*^{+/-} cells alone. Figure 4F has the same issue, and in this graph, the color does not match Figure 4A.

The color-to-genotype relationships for the schematics in Figures 2A and 4E are provided in Figures 1B and 4A, respectively. Due to the high density of germ cells, it is impractical to label each genotype directly in the images. In contrast to Figure 4E, the colors in Figure 4F do not represent genotypes; instead, blue denotes the percentage of SGCs, and red denotes the percentage of germline cysts, as indicated below the bar chart.

(2) The frequencies of bam or bgcn mutant mosaic germaria carrying [wild-type] SGCs or wild-type germ cell cysts with branched fusomes, as well as the average number of wild-type SGCs per germarium and the number of days after heat shock for the representative images, are not provided when Figure 1 is first introduced. Since this is the first time the authors describe these phenotypes, including these details is essential. Without this information, it is difficult for readers to follow and evaluate the presented observations.

Thank you for this constructive suggestion. These quantification data have been added to the revised Figure 1 (Figure 1J, K).

(3) Without the information mentioned in point 2, it causes problems when reading through the section regarding [wild-type] SGCs induced by impairment of differentiation or dedifferentiation. In lines 90-97, the authors use the presence of midbodies between cystocytes as a criterion to determine whether the wild-type GSCs surrounded by tumor GSCs arise through dedifferentiation. However, the cited study (Mathieu et al., 2022) reports that midbodies can be detected between two germ cells within a cyst carrying a branched fusome upon USP8 loss.

Unlike wild-type cystocytes, which undergo incomplete cytokinesis and lack midbodies, those with USP8 loss undergo complete cell division, with the presence of midbodies (white arrow, Figure 1F' from Mathieu et al., 2022) as a marker of the late cytokinesis stage (Mathieu et al., 2022).

(a) Are wild-type germ cell cysts with branched fusomes present in the bam mutant mosaic germlaria? What is the proportion of germlaria containing wild-type SGCs versus those containing wild-type germ cell cysts with branched fusomes?

(b) If all bam mutant mosaic germlaria carry only wild-type GSCs outside the niche and no germlaria contain wild-type germ cell cysts with branched fusomes, then examining midbodies as an indicator of dedifferentiation may not be appropriate.

We appreciate your critical comment. bam mutant mosaic germlaria indeed contained wild-type germline cysts, as evidenced by an SGC frequency of ~70%, rather than 100% (see Figures 2H, 4F, 4J, 6F, 6I, and Figure 6-figure supplement 3C). Since the SGC phenotype depends on the presence of bam or bgcn mutant germline tumors, we quantified it as “the percentage of SGCs relative to the total number of SGCs and germline cysts that are surrounded by germline tumors” (see Lines 103-108). Quantifying the SGC phenotype as “the percentage of germlaria with SGCs” would be imprecise. This is because the presence and number of SGCs were variable among germlaria with bam or bgcn mutant germline clones, and a small number of germlaria entirely lacked these clones. The data of “SGCs per germlarium with both germline clones and out-of-niche wild-type germ cells” have been added to the revised Figure 1 (Figure 1K).

(c) If, however, some germlaria do contain wild-type germ cell cysts with branched fusomes, the authors should provide representative images and quantify their proportion.

Such germlaria could be found in Figure 2G, 3B, 3C, 6D, 6E, and 6H. The percentage of germline cysts can be calculated by “100% - SGC%”.

(d) In line 95, although the authors state that 50 germ cell cysts were analyzed for the presence of midbodies, it would be more informative to specify how many germlaria these cysts were derived from and how many biological replicates were examined.

As noted in our response to points a) and b) above, the germ cells surrounded by germline tumors, rather than germlarial numbers, are more precise for analyzing the phenotype. For this experiment, we examined >50 such germline cysts via confocal microscopy. As the analysis was performed on a defined cellular population, this sample size should be sufficient to support our conclusion.

(4) Note that both bam mutant GSCs and wild-type SGCs can undergo division to generate midbodies (double cells), as shown in Figure 4H. Therefore, the current description of the midbody analysis is confusing. The authors should clarify which cell types were examined and explain how midbodies were interpreted in distinguishing between cell division and differentiation.

We assayed for the presence of midbodies or not specifically within the wild-type germline cysts surrounded by bam or bgcn mutant tumors, not within the tumors themselves (Lines 96-97). As detailed in Lines 90-100, the absence of midbodies was used as a key criterion to exclude the possibility of dedifferentiation.

(5) The data in Figure 5 showing Dpp expression in bam mutant tumorous GSCs are not convincing. The Dpp-lacZ signal appears broadly distributed throughout the germlarium, including in escort cells. To support the claim more clearly, the authors should present

corresponding images for Figures 5D and 5E, in which *dpp* expression was knocked down in the germ cells of *bam* or *bgn* mutant mosaic germaria. Showing these images would help clarify the localization and specificity of *Dpp-lacZ* expression relative to the tumorous GSCs.

Thank you for your constructive comment. RNA in situ hybridization data have been added to support that *bam* or *bgn* mutant germline tumors secrete BMP ligands (Figure 5A-D).

(6) While Figure 6 provides genetic evidence that *bam* mutant tumorous GSCs produce *Dpp* to inhibit the differentiation of wild-type SGCs, it should be noted that these analyses were performed in a *dpp*^{+/−} background. To strengthen the conclusion, the authors should include appropriate controls showing [*dpp*^{+/−}; *bam*^{+/−}] SGCs and [*dpp*^{+/−}; *bam*^{+/−}] germ cell cysts without heat shock (as referenced in Figures 6F and 6I).

Schematic cartoons in Figure 6A and 6B demonstrate that these analyses were performed in a *dpp*^{+/−} background. Figure 6-figure supplement 1 indicates that *dpp*^{+/−} or *gbb*^{+/−} does not affect GSC maintenance, germ cell differentiation, and female fly fertility. Figure 6C is the control for 6D and 6E, and 6G is the control for 6H, with quantification in 6F and 6I. We used *nos>FLP*, not the heat shock method, to induce germline clones in these experiments (see genotypes in Source data 1).

(7) Previous studies have reported that *bam* mutant germ cells cause blunted escort cell protrusions (e.g., Kirilly et al., *Development*, 2011), which are known to contribute to germ cell differentiation (e.g., Chen et al., *Frontiers in Cell and Developmental Biology*, 2022). The authors should include these findings in the Discussion to provide a broader context and to acknowledge how alterations in escort cell morphology may further influence differentiation defects in their model.

Thank you for teaching us! We have included the introduction of these two papers in the revised manuscript (Lines 197-199).

(8) Since fusome morphology is an important readout of SGCs vs differentiation. All the clonal analysis should have fusome staining.

SGC is readily distinguishable from multi-cellular germline cyst based on morphology. In some clonal-analysis experiments, fusome staining was not feasible due to technical limitations such as channel saturation or antibody incompatibility. Thank you for your understanding!

(9) Figure arrangement. It is somewhat difficult to identify the figure panels cited in the text due to the current panel arrangement.

The figure panels were arranged to optimize space while ensuring that related panels are grouped in close proximity for logical comparison. We would be happy to consider any specific suggestions for an alternative layout that could improve clarity.

(10) The number of biological replicates and germaria analyzed should be clearly stated somewhere in the manuscript-ideally in the Methods section or figure legends. Providing this information is essential for assessing data reliability and reproducibility.

The detailed quantification information is labeled directly in figures or described in figure legends, and all raw quantification data are provided in Source data 2.

Reviewer #3 (Public review):

Summary:

Zhang et al. investigated how germline tumors influence the development of neighboring wild-type (WT) germline stem cells (GSC) in the Drosophila ovary. They report that germline tumors inhibit the differentiation of neighboring WT GSCs by arresting them in an undifferentiated state, resulting from reduced expression of the differentiation-promoting factor Bam. They find that these tumor cells produce low levels of the niche-associated signaling molecules Dpp and Gbb, which suppress bam expression and consequently inhibit the differentiation of neighboring WT GSCs non-cell-autonomously. Based on these findings, the authors propose that germline tumors mimic the niche to suppress the differentiation of the neighboring stem cells.

Strengths:

This study addresses an important biological question concerning the interaction between germline tumor cells and WT germline stem cells in the Drosophila ovary. If the findings are substantiated, they could provide valuable insights applicable to other stem cell systems.

We greatly appreciate your valuable comments.

Weaknesses:

Previous work from Xie's lab demonstrated that bam and bgcn mutant GSCs can outcompete WT GSCs for niche occupancy. Furthermore, a large body of literature has established that the interactions between escort cells (ECs) and GSC daughters are essential for proper and timely germline differentiation (the differentiation niche). Disruption of these interactions leads to arrest of germline cell differentiation in a status with weak BMP signaling activation and low bam expression, a phenotype virtually identical to what is reported here. Thus, it remains unclear whether the observed phenotype reflects "direct inhibition by tumor cells" or "arrested differentiation due to the loss of the differentiation niche." Because most data were collected at a very late stage (more than 10 days after clonal induction), when tumor cells already dominate the germarium, this question cannot be solved. To distinguish between these two possibilities, the authors could conduct a time-course analysis to examine the onset of the WT GSC-like single-germ-cell (SGC) phenotype and determine whether early-stage tumor clones with a few tumor cells can suppress the differentiation of neighboring WT GSCs with only a few tumor cells present. If tumor cells indeed produce Dpp and Gbb (as proposed here) to inhibit the differentiation of neighboring germline cells, a small cluster or probably even a single tumor cell generated at an early stage might prevent the differentiation of their neighboring germ cells.

Thank you for your critical comment. The revised manuscript now includes a time-course analysis of the SGC phenotype (Figure 1J). Our data in Figure 6 demonstrate that BMP ligands from germline tumors are required to inhibit SGC differentiation. Furthermore, we have incorporated into the manuscript the possibility that disruption of the differentiation niche may also contribute to the SGC phenotype (Lines 197-199).

The key evidence supporting the claim that tumor cells produce Gpp and Gbb comes from Figures 5 and 6, which suggest that tumor-derived dpp and gbb are required for this inhibition. However, interpretation of these data requires caution. In Figure 5, the authors use dpp-lacZ to support the claim that dpp is upregulated in tumor cells (Figure 5A and 5B). However, the background expression in somatic cells (ECs and pre-follicular cells) differs noticeably between these panels. In Figure 5A, dpp-lacZ expression in somatic cells in 5A is clearly higher than in 5B, and the expression level in tumor cells appears comparable to that in somatic cells (dpp-lacZ single channel). Similarly, in Figure 5B, dpp-lacZ expression in germline cells is also comparable to that in somatic cells.

Providing clear evidence of upregulated dpp and gbb expression in tumor cells (for example, through single-molecular RNA in situ) would be essential.

We greatly appreciate your critical comment. In our data, the expression levels of *dpp-lacZ* in terminal filament and cap cells were highly variable across germaria, even within the same ovary. We have omitted these results in the revised Figure 5. RNA in situ hybridization data have been added to visualize the expression of BMP ligands within *bam* mutant germline tumor cells (Figure 5A-D).

Most tumor data present in this study were collected from the bam[86] null allele, whereas the data in Figure 6 were derived from a weaker bam[BG] allele. This bam[BG] allele is not molecularly defined and shows some genetic interaction with dpp mutants. As shown in Figure 6E, removal of dpp from homozygous bam[BG] mutant leads to germline differentiation (evidenced by a branched fusome connecting several cystocytes, located at the right side of the white arrowhead). In Figure 6D, fusome is likely present in some GFP-negative bam[BG]/bam[BG] cells. To strengthen their claim that the tumor produces Dpp and Gbb to inhibit WT germline cell differentiation, the authors should repeat these experiments using the bam[86] null allele.

Although a structure resembling a "branched fusome" is visible in Figure 6E (right of the white arrowhead), it is an artifact resulting from the cytoplasm of GFP-positive follicle cells, which also stain for α -Spectrin, projecting between germ cells of different clones (see the merged image). In both our previous (Zhang et al., 2023) and current studies, *bam*^{BG} was functionally indistinguishable from *bam*^{A86} in its ability to block GSC differentiation and induce the SGC phenotype (Figure 1J). Given this, we believe that repeating the extensive experiments in Figure 6 with the *bam*^{A86} allele would be scientifically redundant and would not change the key conclusion of our study.

It is well established that the stem niche provides multiple functional supports for maintaining resident stem cells, including physical anchorage and signaling regulation. In Drosophila, several signaling molecules produced by the niche have been identified, each with a distinct function - some promoting stemness, while others regulate differentiation. Expression of Dpp and Gbb alone does not substantiate the claim that these tumor cells have acquired the niche-like property. To support their assertion that these tumors mimic the niche, the authors should provide additional evidence showing that these tumor cells also express other niche-associated markers. Alternatively, they could revise the manuscript title to more accurately reflect their findings.

Dpp and Gbb are the key niche signals from cap cells for maintaining GSC stemness. Our work demonstrates that germline tumors can specifically mimic this signaling function, not the full suite of cap cell properties, to create a non-cell-autonomous differentiation block. The current title "Tumors mimic the niche to inhibit neighboring stem cell differentiation" reflects this precise concept: a partial, functional mimicry of the niche's most relevant activity in this context. We feel it is an appropriate and compelling summary of our main conclusion.

In the Method section, the authors need to provide details on how dpp-lacZ expression levels were quantified and normalized.

Because of the highly variable expression levels in terminal filament and cap cells, we have omitted the *dpp-lacZ* results in the revised manuscript.

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

Minor points

(1) Not all readers may be familiar with the *nos>FLP/FRT* or *hs-FLP/FRT* systems. It would be helpful if the authors could briefly introduce these genetic mosaic systems and explain how they were used in this study before presenting the results.

Thank you for this constructive suggestion. Such brief introduction has been added to the revised manuscript (Lines 64-70).

(2) Line 68-70: "Surprisingly, ...outside the niche retained a GSC-like single-germ-cell (SGC) morphology, even when encapsulated within egg chambers (Figure 1C, D, Figure 1-figure supplement 1).

(3) The figure citation is not appropriate, as Figures 1C and 1D do not show "single germ cells (SGCs) encapsulated within egg chambers." To improve clarity, the authors could revise the sentence as follows: "Surprisingly, wild-type germ cells located outside the niche retained a GSC-like single-germ-cell (SGC) morphology (Figures 1C and D), even when encapsulated within egg chambers (Figure 1-figure supplement 1)." This modification would make the description consistent with the figure content and easier for readers to follow.

Thank you for teaching us! The manuscript has been revised following this suggestion (Lines 70-73).

(4) Line 106-110. The description is confusing. The authors state, "Under normal conditions... Notably, 74% of SGCs ($n = 132$) were GFP-negative, while the remaining 26% were GFP-positive (Figure 2B, C). However, Figure 2B shows the *bam* mutant mosaic germlaria, and Figure 2C does not specify the genotypes of the germlaria used for the analysis of GSCs, CBs, and SGCs. The authors should clarify the experimental conditions and genotypes corresponding to each panel. In addition, it would be more informative to indicate how many germlaria these quantified GSCs, CBs, and SGCs were derived from.

(5) Throughout the manuscript, the authors report the number of SGCs analyzed (e.g., Lines 149-151). However, it would be more informative to also indicate how many germlaria these quantified SGCs were derived from. Providing this information would help readers assess the sampling size and variability across biological replicates.

Thank you for your suggestion. As shown in Figure 2B, these wild-type (RFP-positive) GSCs and CBs were also derived from *bam* mutant mosaic germlaria. The phrase "under normal conditions" has been deleted from the revised manuscript to prevent any potential ambiguity. Given the specificity of the SGC phenotype, the germ cells surrounded by germline tumors, rather than germlaria numbers, are more precise for its quantification (Lines 103-108). The data of "SGCs per germlarium with both germline clones and out-of-niche wild-type germ cells" have been added to the revised Figure 1K.

Reviewer #3 (Recommendations for the authors):

(1) Additionally, the authors should clarify what the "red dot" signal in the GFP-positive cap cell in Figure 3 F (left panel) represents.

The "red dot" is an asterisk that is used to mark a cap cell (Line 620).

(2) Finally, on line 266, "*bamP*-GFP-positive" should be corrected to "*bamP*-GFP-negative."

It should be "*bamP*-GFP-positive", not "*bamP*-GFP-negative" (see Figure 2B).

Reference:

Mathieu, J., Michel-Hissier, P., Boucherit, V., and Huynh, J.R. (2022). The deubiquitinase USP8 targets ESCRT-III to promote incomplete cell division. *Science* 376, 818-823.

Zhang, Q., Zhang, Y., Zhang, Q., Li, L., and Zhao, S. (2023). Division promotes adult stem cells to perform active niche competition. *Genetics* 224.

Zhao, S., Fortier, T.M., and Baehrecke, E.H. (2018). Autophagy Promotes Tumor-like Stem Cell Niche Occupancy. *Curr Biol* 28, 3056-3064.e3053.

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